



US012414949B2

(12) **United States Patent**
Flaherty et al.

(10) **Patent No.:** **US 12,414,949 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **PYRAZOLYL PYRIMIDINONE COMPOUNDS AND THE USES THEREOF**

(71) Applicant: **Purdue Research Foundation**, West Lafayette, IN (US)

(72) Inventors: **Daniel P. Flaherty**, West Lafayette, IN (US); **Richard M. van Rijn**, West Lafayette, CA (US); **Val J. Watts**, West Lafayette, IN (US); **Jason A Scott**, West Lafayette, IN (US)

(73) Assignee: **Purdue Research Foundation**, West Lafayette, IN (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 8 days.

(21) Appl. No.: **18/200,419**

(22) Filed: **May 22, 2023**

(65) **Prior Publication Data**
US 2023/0293523 A1 Sep. 21, 2023

Related U.S. Application Data

(62) Division of application No. 17/063,747, filed on Oct. 6, 2020, now Pat. No. 11,690,844.

(60) Provisional application No. 62/911,604, filed on Oct. 7, 2019.

(51) **Int. Cl.**
A61K 31/506 (2006.01)
C07D 403/04 (2006.01)

(52) **U.S. Cl.**
CPC **A61K 31/506** (2013.01); **C07D 403/04** (2013.01)

(58) **Field of Classification Search**
CPC A61K 31/506; C07D 403/04
USPC 514/273
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,662,176 B2 5/2020 Watts et al.
11,667,623 B2 * 6/2023 Watts et al.
2019/0023688 A1 * 1/2019 Watts et al.

FOREIGN PATENT DOCUMENTS

WO 2012/154880 11/2012
WO WO 2012/154880 A1 * 11/2013

OTHER PUBLICATIONS

Ballell, et. al., ChemMedChem (2013), 8(2), 313-321. (Year: 2013).
Guo et al., "Adenylyl cyclase Rv1625c of Mycobacterium tuberculosis: a progenitor of mammalian adenylyl cyclases" EMBO J, Jul. 16, 2001;20(14):3667-3675. doi: 10.1093/emboj/20.14.3667 (Year: 2001).*

Wong et al. "Cyclic AMP Is A Critical Mediator Of Intrinsic Drug Resistance And Fatty Acid Metabolism In M. Tuberculosis" Research Article, Microbiology and Infectious Disease, eLife 2023, <https://doi.org/10.7554/eLife.81177> (Year: 2023).*

Al-Hasani et al., "Molecular Mechanisms of Opioid Receptor-Dependent Signaling and Behavior", Anesthesiology, Dec. 2011, 115(6):1363-1381, doi:10.1097/ALN.0b013e318238bba6 (Year: 2011).*

U.S. Appl. No. 11/667,623, filed Jun. 6, 2023, Watts.*
Dessauer et al., "International Union of Basic and Clinical Pharmacology. CI. Structures and Small Molecule Modulators of Mammalian Adenylyl Cyclases", Pharmacological Reviews, Apr. 2017, 69(2), 93-139; DOI: <https://doi.org/10.1124/pr.116.013078>; (Year: 2017).*

Li et al. "Cyclic Nucleotide Signaling In Sensory Neuron Hyperexcitability And Chronic Pain After Nerve Injury" Neurobiol Pain Mar. 8, 2019;6:100028. doi: 10.1016/j.ynpai.2019.100028; (Year: 2019).*

Hughes et al., "Principles of Early Drug Discovery", Br J Pharmacol Mar. 2011;162(6):1239-1249. doi: 10.1111/j.1476-5381.2010.01127 (Year: 2011).*

Simonds "G protein regulation of adenylyl cyclase", Trends Pharmacol Sci, Feb. 1999;20(2):66-73. doi: 10.1016/s0165-6147(99) 01307-3. (Year: 1999).*

Madoux, et. al., Journal of Biomolecular Screening (2010), 15(8), 907-917 (Year: 2010).*

Gamble et al. "Mu-Opioid Receptor And Receptor Tyrosine Kinase Crosstalk: Implications In Mechanisms Of Opioid Tolerance, Reduced Analgesia To Neuropathic Pain, Dependence & Reward", Review article, Front. Syst. Neurosci., Nov. 30, 2022, vol. 16, <https://doi.org/10.3389/fnsys.2022.1059089> (Teaching Ref.) (Year: 2022).*

Johnson et al., "Chemical activation of adenylyl cyclase Rv1625c inhibits growth of Mycobacterium tuberculosis on cholesterol and modulates intramacrophage signaling", Mol Microbiol Jul. 2017; 105(2):294-308. doi: 10.1111/mmi.13701. Epub May 23, 2017) (Year: 2017).*

Sheikhpour et al., "Gene Expression and In Vitro Pharmacogenetic Studies of Dopamine and Serotonin Gene Receptors in Tuberculosis" 1 Tanaffos Feb. 2021;20(2):126-133; (Year: 2021).*

Shah et al., Toll-like Receptor-Dependent Negative Effects of Opioids: A Battle between Analgesia and Hyperalgesia Front Immunol May 31, 2017;8:642. doi: 10.3389/fimmu.2017.00642; (Year: 2017).*

Lee et al., "Pain And Its Clinical Associations In Individuals With Cystic Fibrosis A Systematic Review, Chron Respir Dis. Feb. 12, 2016;13(2):102-117. doi: 10.1177/1479972316631135" (Year: 2016).*

Efendiev et al. AKAPs and Adenylyl Cyclase in Cardiovascular Physiology and Pathology', J Cardiovasc Pharmacol. Oct. 2011;58(4):339-344. doi: 10.1097/FJC.0b013e31821bc3f0. (Year: 2011).*

Kang et al., "Analgesic effects of adenylyl cyclase inhibitor NB001 on bone cancer pain in a mouse model", Mol Pain. Sep. 9, 2016; doi: 10.1177/1744806916652409. (Year: 2016).*

Volkow, N., et al., "Opioid Abuse in Chronic Pain—Misconceptions and Mitigation Strategies", New Engl J Med, 374, pp. 1253-1263, 2016.

(Continued)

Primary Examiner — Jeffrey H Murray

Assistant Examiner — Grace Ching Hsu

(74) *Attorney, Agent, or Firm* — Purdue Research Foundation

(57) **ABSTRACT**

The present invention relates to a method of treatment for chronic pain, opioid dependence, alcohol use disorder or autism using a class of pyrimidinone compounds, an adenylyl cyclase 1 (AC1) inhibitor. The invention described herein also pertains to pharmaceutical compositions and methods for treating diseases in mammals using those compounds disclosed herein.

3 Claims, No Drawings

(56)

References Cited

OTHER PUBLICATIONS

Skolnick, P., et al., "Re-energizing the Development of Pain Therapeutics in Light of the Opioid Epidemic", *Neuron* 92, pp. 294-297, 2016.

Zhuo, M., et al., "Targeting neuronal adenylyl cyclase for the treatment of chronic pain", *Drug Discov Today*, vol. 17, 17, pp. 573-582, 2012.

Sadana, R., et al., "Physiological Roles for G Protein-Regulated Adenylyl Cyclase Isoforms: Insights from Knockout and Overexpression Studies", *Neurosignals* 17, pp. 5-22, 2008.

Sanabra, C., et al., "Neuroanatomical distribution and neurochemical characterization of cells expressing adenylyl cyclase isoforms in mouse and rat brain", *J Chem Neuroanat*, 41, pp. 43-54, 2011.

Bosse, K., et al., "Adenylyl Cyclase 1 Is Required for Ethanol-Induced Locomotor Sensitization and Associated Increases in NMDA Receptor Phosphorylation and Function in the Dorsal Medial Striatum", *J. Pharmacol. Exp. Ther.*, 363, pp. 148-155, 2017.

Sethna, F., et al. "Enhanced expression of ADCY1 underlies aberrant neuronal signalling and behaviour in a syndromic autism model", *Nat. Commun.*, 14359, pp. 11, 2017.

Baell, J. et al., "New Substructure Filters for Removal of Pan Assay Interference Compounds (PAIS) from Screening Libraries and for Their Exclusion in Bioassays", *J. Med. Chem.*, 53, pp. 2719-2740, 2010.

Cumbay, M., et al., "Heterologous Sensitization of Recombinant Adenylate Cyclases by Activation of D2 Dopamine Receptors", *J. Pharmacol Exp Ther.*, 297, pp. 1201-1399, 2001.

Masada, N., et al., "Distinct Mechanisms of Calmodulin Binding and Regulation of Adenylyl Cyclases 1 and 8", *Biochemistry*, 51, pp. 7917-7929, 2012.

Conley, J., et al., "Drug-induced Sensitization of Adenylyl Cyclase: Assay Streamlining and Miniaturization for Small Molecule and siRNA Screening Applications", *J Vis Exp*, e51218, pp. 10, 2014.

Brust, T., et al., "Bias Analyses of Preclinical and Clinical D2 Dopamine Ligands: Studies with Immediate and Complex Signaling Pathways", *J. Pharmacol. Exp. Ther.*, 352, pp. 480-493, 2015.

Conley J., et al., "Development of a High-Throughput Screening Paradigm for the Discovery of Small-Molecule Modulators of Adenylyl Cyclase: Identification of an Adenylyl Cyclase 2 Inhibitor", *J. Pharmacol. Exp. Ther.*, 347, pp. 276-287, 2013.

Brust, T., et al., "New functional activity of aripiprazole revealed: Robust antagonism of D2 dopamine receptor—stimulated Gbg signaling", *Biochem. Pharmacol.*, 93, pp. 85-91, 2015.

Corder, G., et al., "Constitutive m-Opioid Receptor Activity Leads to Long-Term Endogenous Analgesia and Dependence", *Science*, 341, pp. 1394-1399, 2013.

Van Rijn, R., et al., "Emergence of Functional Spinal Delta Opioid Receptors After Chronic Ethanol Exposure", *Biol. Psychiatry*, 71, pp. 232-238, 2012.

Due-Hansen, M., et al., "A protocol for amide bond formation with electron deficient amines and sterically hindered substrates", *Org. Biomol. Chem.*, 14, pp. 430-433, 2016.

Maddoux, F. et al., "An Ultra-High Throughput Cell-Based Screen for Wee1 Degradation Inhibitors," *Journal of Biomolecular Screening*, 15(8), pp. 907-917, 2010.

Ballell, L. et al., "Fueling Open-Source Drug Discovery: 177 Small-Molecule Leads against Tuberculosis," *ChemMedChem*, 8(2), pp. 313-321, 2013.

* cited by examiner

PYRAZOLYL PYRIMIDINONE COMPOUNDS AND THE USES THEREOF

CROSS REFERENCE TO RELATED APPLICATIONS

This present patent application is a divisional application of U.S. patent application Ser. No. 17/063,747, filed Oct. 6, 2020, which relates to and claims the priority benefit of U.S. Provisional Application Ser. No. 62/911,604, filed Oct. 7, 2019, the contents each of which is hereby incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present invention relates to a method of treatment for chronic pain, opioid dependence, alcohol use disorder or autism. Particularly, the present disclosure relates to a method of treatment for chronic pain, opioid dependence, alcohol use disorder or autism using a class of pyrazolyl pyrimidinone adenylyl cyclase inhibitors. The invention described herein also pertains to pharmaceutical compositions and methods for treating diseases in mammals using compounds disclosed herein.

BACKGROUND

This section introduces aspects that may help facilitate a better understanding of the disclosure. Accordingly, these statements are to be read in this light and are not to be understood as admissions about what is or is not prior art.

Chronic pain is a major health concern that costs the US more than \$635 billion per year (Gaskin and Richard, (2012) *J. Pain* 13:715-724). In addition to the financial impact, patients with chronic pain suffer extreme physical, emotional, and social burdens. For example, individuals often become socially isolated and confined to home as a result of their chronic pain that is not well-controlled by today's available treatments. The drugs used for the management of chronic pain include opioid analgesics, neuronal stabilizers such as anticonvulsants, and antidepressants. Opioids are the most widely used, and a recent NIH report indicates that there are significant problems associated with long-term opioid therapy for chronic pain (Volkow and McLellan (2016) *N Engl J Med* 374:1253-1263.). None of the agents provide sufficient relief to allow patients to return to their normal activity level. Moreover, current pharmaceutical industry has retreated from studying novel pain therapeutics due to the enormous risk (Skolnick and Volkow (2016) *Neuron* 92:294-297.). These observations indicate an essential need to identify new agents acting on unique targets in the war on chronic pain.

Neurobiological, genetic, and preclinical studies have implicated neuronal adenylyl cyclase type I (AC1) as a potential new drug target (Zhuo (2012) *Drug Discov Today* 17:573-582.). Adenylyl cyclases (AC) are an enzyme family that serve as effectors of numerous G protein coupled receptors (e.g. opioid and dopamine receptors) and produce the second messenger cAMP from ATP. Nine membrane-bound isoforms of AC share a similar structure that includes an intracellular N-terminus, followed by two membrane-spanning domains alternating with two cytoplasmic (catalytic) domains that can be further divided into a and b regions (Sadana and Dessauer, (2009) *Neurosignals* 17:5-22.). The C1a and C2a domains make up the catalytic portion of the enzyme, and an X-ray crystal structure with the C1a domain from AC5 and the C2a domain from AC2

was solved in 1997 (Tesmer et al., (1997) *Science* 278:1907-1916.). In contrast, no structural information exists regarding N-terminus, C1b, or C2b domains for any isoform. Each isoform is uniquely regulated by G protein α and $\beta\gamma$ subunits, Ca^{2+} , protein kinases, posttranslational modifications, and subcellular localization (Willoughby and Cooper, (2007) *Physiol Rev* 87:965-1010).

Group 1 ACs, represented by AC1, AC3, and AC8, are stimulated by calmodulin in a Ca^{2+} -dependent manner. Group 2 ACs are characterized by their conditional stimulation by $\text{G}\beta\gamma$ subunits and are represented by AC2, AC4, and AC7. AC2 and AC7 are also activated by protein kinase C. Group 3 ACs include AC5 and AC6, show robust negative regulation by $\text{G}\alpha_i$ subunits, and are also inhibited by sub-micromolar concentration of Ca^{2+} as well as protein kinase A. Group 4 ACs contains only one member, AC9, which is unique among the ACs in being relatively insensitive to activation by the small molecule diterpene, forskolin.

Membrane-bound ACs are highly expressed in the central nervous system and generally have overlapping expression patterns (Sanabra and Mengod, (2011). *J Chem Neuroanat* 41:43-54.). Multiple AC isoforms are typically expressed in individual cell types, making it difficult to elucidate the function(s) of individual isoforms in either native tissues or cell lines. This problem has been addressed using a variety of recombinant approaches, including overexpression, site-directed mutagenesis, and, most notably, global genetic deletions. These animals lacking one or multiple AC isoforms have been essential tools to inform on the physiological roles of AC signaling in the central nervous system (Sadana R et al., (2009). *Neurosignals* 17:5-22).

Physiological roles of AC1 and AC8: AC1 and AC8 are robustly activated by Ca^{2+} /calmodulin (Ca^{2+} /CaM) and have overlapping expression patterns in neuronal tissues, including the hippocampus and several cortical regions (Defer N, et al., (2000). *Am J Physiol Renal Physiol* 279:F400-F416). To explore their relative physiological roles, a number of studies have been carried out with mice lacking either AC1 ($\text{AC1}^{-/-}$), AC8 ($\text{AC8}^{-/-}$), or both isoforms (double knock out mice, DKO). Initial experiments with animals lacking Ca^{2+} /CaM-stimulated cyclases focused on long-term memory (LTM) and long-term potentiation (LTP) due to their high level of expression in the hippocampus. The results of these experiments implicated AC1 and AC8 in LTP and LTM (Ferguson and Storm, 2004). Importantly, it was found that long lasting LTP and memory deficits were marked in animals lacking both AC1 and AC8 (DKO mice), but were mostly absent in animals deficient in only a single AC isoform. However, a few studies found that $\text{AC1}^{-/-}$ mice showed modest deficits in other forms of LTP (Chen et al., (2014), *Mol Pain* 10:65.), including a reduction in remote contextual fear memory that was only observed at a single time point. These observations clearly implicate Ca^{2+} /CaM-stimulated cyclases in LTP and certain models of memory; however, selectively targeting a single AC isoform markedly reduces the overall deficits. Furthermore, these findings also emphasize the benefits of pharmacologically targeting overactive AC1 in dose-dependent fashion versus complete inhibition or genetic deletion.

Additionally, AC1 knock out mice show less reward when given opioids and show reduced symptoms of opioid dependence during withdrawal. Additional reports suggest that AC1 inhibition may also provide a useful therapeutic intervention for alcohol use disorder and autism (Bosse K E et al., *J. Pharmacol. Exp. Ther.* 2017, 363 (2) 148-155; Sethna F., et al. *Nat. Commun.* 2017, 8, 14359).

Unfortunately, until now, the selective inhibition of ACs has not been achieved, and simultaneous inhibition of multiple adenylyl cyclase isoforms would likely result in significant adverse effects. There are unmet needs for better and safer medications targeting adenylyl cyclases for various therapeutic uses, including pain, opioid dependence, alcohol use disorder and autism.

DETAILED DESCRIPTION

While the concepts of the present disclosure are illustrated and described in detail in the figures and the description herein, results in the figures and their description are to be considered as exemplary and not restrictive in character; it being understood that only the illustrative embodiments are shown and described and that all changes and modifications that come within the spirit of the disclosure are desired to be protected.

As used herein, the following terms and phrases shall have the meanings set forth below. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art.

In the present disclosure the term “about” can allow for a degree of variability in a value or range, for example, within 10%, within 5%, or within 1% of a stated value or of a stated limit of a range. In the present disclosure, the term “substantially” can allow for a degree of variability in a value or range, for example, within 90%, within 95%, 99%, 99.5%, 99.9%, 99.99%, or at least about 99.999% or more of a stated value or of a stated limit of a range.

The term “substituted” as used herein refers to a functional group in which one or more hydrogen atoms contained therein are replaced by one or more non-hydrogen atoms. The term “functional group” or “substituent” as used herein refers to a group that can be or is substituted onto a molecule. Examples of substituents or functional groups include, but are not limited to, a halogen (e.g., F, Cl, Br, and I); an oxygen atom in groups such as hydroxyl groups, alkoxy groups, aryloxy groups, aralkyloxy groups, oxo (carbonyl) groups, carboxyl groups including carboxylic acids, carboxylates, and carboxylate esters; a sulfur atom in groups such as thiol groups, alkyl and aryl sulfide groups, sulfoxide groups, sulfone groups, sulfonyl groups, and sulfonamide groups; a nitrogen atom in groups such as amines, azides, hydroxylamines, cyano, nitro groups, N-oxides, hydrazides, and enamines; and other heteroatoms in various other groups.

The term “alkyl” as used herein refers to substituted or unsubstituted straight chain and branched alkyl groups and cycloalkyl groups having from 1 to about 20 carbon atoms (C_1 - C_{20}), 1 to 12 carbons (C_1 - C_{12}), 1 to 8 carbon atoms (C_1 - C_8), or, in some embodiments, from 1 to 6 carbon atoms (C_1 - C_6). Examples of straight chain alkyl groups include those with from 1 to 8 carbon atoms such as methyl, ethyl, n-propyl, n-butyl, n-pentyl, n-hexyl, n-heptyl, and n-octyl groups. Examples of branched alkyl groups include, but are not limited to, isopropyl, iso-butyl, sec-butyl, t-butyl, neopentyl, isopentyl, and 2,2-dimethylpropyl groups. As used herein, the term “alkyl” encompasses n-alkyl, isoalkyl, and anteisoalkyl groups as well as other branched chain forms of alkyl. Representative substituted alkyl groups can be substituted one or more times with any of the groups listed herein, for example, amino, hydroxy, cyano, carboxy, nitro, thio, alkoxy, and halogen groups.

The term “alkenyl” as used herein refers to substituted or unsubstituted straight chain and branched divalent alkenyl

and cycloalkenyl groups having from 2 to 20 carbon atoms (C_2 - C_{20}), 2 to 12 carbons (C_2 - C_{12}), 2 to 8 carbon atoms (C_2 - C_8) or, in some embodiments, from 2 to 4 carbon atoms (C_2 - C_4) and at least one carbon-carbon double bond. Examples of straight chain alkenyl groups include those with from 2 to 8 carbon atoms such as $-\text{CH}=\text{CH}-$, $-\text{CH}=\text{CHCH}_2-$, and the like. Examples of branched alkenyl groups include, but are not limited to, $-\text{CH}=\text{C}(\text{CH}_3)-$ and the like.

An alkynyl group is the fragment, containing an open point of attachment on a carbon atom that would form if a hydrogen atom bonded to a triply bonded carbon is removed from the molecule of an alkyne. The term “hydroxyalkyl” as used herein refers to alkyl groups as defined herein substituted with at least one hydroxyl ($-\text{OH}$) group.

The term “cycloalkyl” as used herein refers to substituted or unsubstituted cyclic alkyl groups such as, but not limited to, cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, and cyclooctyl groups. In some embodiments, the cycloalkyl group can have 3 to about 8-12 ring members, whereas in other embodiments the number of ring carbon atoms range from 3 to 4, 5, 6, or 7. In some embodiments, cycloalkyl groups can have 3 to 6 carbon atoms (C_3 - C_6). Cycloalkyl groups further include polycyclic cycloalkyl groups such as, but not limited to, norbornyl, adamantyl, bornyl, camphenyl, isocamphenyl, and carenyl groups, and fused rings such as, but not limited to, decaliny, and the like.

The term “acyl” as used herein refers to a group containing a carbonyl moiety wherein the group is bonded via the carbonyl carbon atom. The carbonyl carbon atom is also bonded to another carbon atom, which can be part of a substituted or unsubstituted alkyl, aryl, aralkyl cycloalkyl, cycloalkylalkyl, heterocyclyl, heterocyclalkyl, heteroaryl, heteroarylalkyl group or the like. In the special case wherein the carbonyl carbon atom is bonded to a hydrogen, the group is a “formyl” group, an acyl group as the term is defined herein. An acyl group can include 0 to about 12-40, 6-10, 1-5 or 2-5 additional carbon atoms bonded to the carbonyl group. An acryloyl group is an example of an acyl group. An acyl group can also include heteroatoms within the meaning here. A nicotinoyl group (pyridyl-3-carbonyl) is an example of an acyl group within the meaning herein. Other examples include acetyl, benzoyl, phenylacetyl, pyridylacetyl, cinnamoyl, and acryloyl groups and the like. When the group containing the carbon atom that is bonded to the carbonyl carbon atom contains a halogen, the group is termed a “haloacyl” group. An example is a trifluoroacetyl group.

The term “aryl” as used herein refers to substituted or unsubstituted cyclic aromatic hydrocarbons that do not contain heteroatoms in the ring. Thus aryl groups include, but are not limited to, phenyl, azulenyl, heptalenyl, biphenyl, indacenyl, fluorenyl, phenanthrenyl, triphenylenyl, pyrenyl, naphthacenyl, chrysenyl, biphenylenyl, anthracenyl, and naphthyl groups. In some embodiments, aryl groups contain about 6 to about 14 carbons (C_6 - C_{14}) or from 6 to 10 carbon atoms (C_6 - C_{10}) in the ring portions of the groups. Aryl groups can be unsubstituted or substituted, as defined herein. Representative substituted aryl groups can be mono-substituted or substituted more than once, such as, but not limited to, 2-, 3-, 4-, 5-, or 6-substituted phenyl or 2-8 substituted naphthyl groups, which can be substituted with carbon or non-carbon groups such as those listed herein.

The term “aralkyl” and “arylalkyl” as used herein refers to alkyl groups as defined herein in which a hydrogen or carbon bond of an alkyl group is replaced with a bond to an aryl group as defined herein. Representative aralkyl groups

5

include benzyl and phenylethyl groups and fused (cycloalkylaryl)alkyl groups such as 4-ethyl-indanyl. Aralkenyl groups are alkenyl groups as defined herein in which a hydrogen or carbon bond of an alkyl group is replaced with a bond to an aryl group as defined herein.

The term "heterocyclyl" as used herein refers to substituted or unsubstituted aromatic and non-aromatic ring compounds containing 3 or more ring members, of which, one or more is a heteroatom such as, but not limited to, B, N, O, and S. Thus, a heterocyclyl can be a cycloheteroalkyl, or a heteroaryl, or if polycyclic, any combination thereof. In some embodiments, heterocyclyl groups include 3 to about 20 ring members, whereas other such groups have 3 to about 15 ring members. In some embodiments, heterocyclyl groups include heterocyclyl groups that include 3 to 8 carbon atoms (C_3-C_8), 3 to 6 carbon atoms (C_3-C_6) or 6 to 8 carbon atoms (C_6-C_8).

A heteroaryl ring is an embodiment of a heterocyclyl group. The phrase "heterocyclyl group" includes fused ring species including those that include fused aromatic and non-aromatic groups. Representative heterocyclyl groups include, but are not limited to pyrrolidinyl, azetidiny, piperidinyl, piperazinyl, morpholinyl, chromanyl, indolinonyl, isoindolinonyl, furanyl, pyrrolidinyl, pyridinyl, pyrazinyl, pyrimidinyl, triazinyl, thiophenyl, tetrahydrofuranyl, pyrrolyl, oxazoly, oxadiazoly, imidazoly, triazyoly, tetrazoly, benzoxazoliny, benzthiazoliny, and benzimidazoliny groups.

The term "heterocyclylalkyl" as used herein refers to alkyl groups as defined herein in which a hydrogen or carbon bond of an alkyl group as defined herein is replaced with a bond to a heterocyclyl group as defined herein. Representative heterocyclylalkyl groups include, but are not limited to, furan-2-yl methyl, furan-3-yl methyl, pyridine-3-yl methyl, tetrahydrofuran-2-yl methyl, and indol-2-yl propyl.

The term "heteroarylalkyl" as used herein refers to alkyl groups as defined herein in which a hydrogen or carbon bond of an alkyl group is replaced with a bond to a heteroaryl group as defined herein.

The term "alkoxy" as used herein refers to an oxygen atom connected to an alkyl group, including a cycloalkyl group, as are defined herein. Examples of linear alkoxy groups include but are not limited to methoxy, ethoxy, propoxy, butoxy, pentyloxy, hexyloxy, and the like. Examples of branched alkoxy include but are not limited to isopropoxy, sec-butoxy, tert-butoxy, isopentyloxy, isohexyloxy, and the like. Examples of cyclic alkoxy include but are not limited to cyclopropyloxy, cyclobutyloxy, cyclopentyloxy, cyclohexyloxy, and the like. An alkoxy group can further include double or triple bonds, and can also include heteroatoms. For example, an allyloxy group is an alkoxy group within the meaning herein. A methoxyethoxy group is also an alkoxy group within the meaning herein, as is a methylenedioxy group in a context where two adjacent atoms of a structure are substituted therewith.

The term "amine" as used herein refers to primary, secondary, and tertiary amines having, e.g., the formula $N(\text{group})_3$ wherein each group can independently be H or non-H, such as alkyl, aryl, and the like. Amines include but are not limited to $R-NH_2$, for example, alkylamines, arylamines, alkylaryl amines; R_2NH wherein each R is independently selected, such as dialkylamines, diarylamines, aralkylamines, heterocyclylamines and the like; and R_3N wherein each R is independently selected, such as trialkylamines, dialkylaryl amines, alkyl diarylamines, triaryl amines, and the like. The term "amine" also includes ammonium ions as used herein.

6

The term "amino group" as used herein refers to a substituent of the form $-NH_2$, $-NHR$, $-NR_2$, $-NR_3^+$, wherein each R is independently selected, and protonated forms of each, except for $-NR_3^+$, which cannot be protonated. Accordingly, any compound substituted with an amino group can be viewed as an amine. An "amino group" within the meaning herein can be a primary, secondary, tertiary, or quaternary amino group. An "alkylamino" group includes a mono alkylamino, dialkylamino, and trialkylamino group.

The terms "halo," "halogen," or "halide" group, as used herein, by themselves or as part of another substituent, mean, unless otherwise stated, a fluorine, chlorine, bromine, or iodine atom.

The term "haloalkyl" group, as used herein, includes mono-halo alkyl groups, poly-halo alkyl groups wherein all halo atoms can be the same or different, and per-halo alkyl groups, wherein all hydrogen atoms are replaced by halogen atoms, such as fluoro. Examples of haloalkyl include trifluoromethyl, 1,1-dichloroethyl, 1,2-dichloroethyl, 1,3-dibromo-3,3-difluoropropyl, perfluorobutyl, $-CF(CH_3)_2$ and the like.

The term "optionally substituted," or "optional substituents," as used herein, means that the groups in question are either unsubstituted or substituted with one or more of the substituents specified. When the groups in question are substituted with more than one substituent, the substituents may be the same or different. When using the terms "independently," "independently are," and "independently selected from" mean that the groups in question may be the same or different. Certain of the herein defined terms may occur more than once in the structure, and upon such occurrence each term shall be defined independently of the other.

The compounds described herein may contain one or more chiral centers, or may otherwise be capable of existing as multiple stereoisomers. It is to be understood that in one embodiment, the invention described herein is not limited to any particular stereochemical requirement, and that the compounds, and compositions, methods, uses, and medicaments that include them may be optically pure, or may be any of a variety of stereoisomeric mixtures, including racemic and other mixtures of enantiomers, other mixtures of diastereomers, and the like. It is also to be understood that such mixtures of stereoisomers may include a single stereochemical configuration at one or more chiral centers, while including mixtures of stereochemical configuration at one or more other chiral centers.

Similarly, the compounds described herein may include geometric centers, such as cis, trans, E, and Z double bonds. It is to be understood that in another embodiment, the invention described herein is not limited to any particular geometric isomer requirement, and that the compounds, and compositions, methods, uses, and medicaments that include them may be pure, or may be any of a variety of geometric isomer mixtures. It is also to be understood that such mixtures of geometric isomers may include a single configuration at one or more double bonds, while including mixtures of geometry at one or more other double bonds.

As used herein, the term "salts" and "pharmaceutically acceptable salts" refer to derivatives of the disclosed compounds wherein the parent compound is modified by making acid or base salts thereof. Examples of pharmaceutically acceptable salts include, but are not limited to, mineral or organic acid salts of basic groups such as amines; and alkali or organic salts of acidic groups such as carboxylic acids. Pharmaceutically acceptable salts include the conventional

non-toxic salts or the quaternary ammonium salts of the parent compound formed, for example, from non-toxic inorganic or organic acids. For example, such conventional non-toxic salts include those derived from inorganic acids such as hydrochloric, hydrobromic, sulfuric, sulfamic, phosphoric, and nitric; and the salts prepared from organic acids such as acetic, propionic, succinic, glycolic, stearic, lactic, malic, tartaric, citric, ascorbic, pantoic, maleic, hydroxymaleic, phenylacetic, glutamic, benzoic, salicylic, sulfanilic, 2-acetoxybenzoic, fumaric, toluenesulfonic, methanesulfonic, ethane disulfonic, oxalic, and isethionic, and the like.

Pharmaceutically acceptable salts can be synthesized from the parent compound which contains a basic or acidic moiety by conventional chemical methods. In some instances, such salts can be prepared by reacting the free acid or base forms of these compounds with a stoichiometric amount of the appropriate base or acid in water or in an organic solvent, or in a mixture of the two; generally, nonaqueous media like ether, ethyl acetate, ethanol, isopropanol, or acetonitrile are preferred. Lists of suitable salts are found in Remington's Pharmaceutical Sciences, 17th ed., Mack Publishing Company, Easton, Pa., 1985, the disclosure of which is hereby incorporated by reference.

The term "solvate" means a compound, or a salt thereof, that further includes a stoichiometric or non-stoichiometric amount of solvent bound by non-covalent intermolecular forces. Where the solvent is water, the solvate is a hydrate.

Further, in each of the foregoing and following embodiments, it is to be understood that the formulae include and represent not only all pharmaceutically acceptable salts of the compounds, but also include any and all hydrates and/or solvates of the compound formulae or salts thereof. It is to be appreciated that certain functional groups, such as the hydroxy, amino, and like groups form complexes and/or coordination compounds with water and/or various solvents, in the various physical forms of the compounds. Accordingly, the above formulae are to be understood to include and represent those various hydrates and/or solvates. In each of the foregoing and following embodiments, it is also to be understood that the formulae include and represent each possible isomer, such as stereoisomers and geometric isomers, both individually and in any and all possible mixtures. In each of the foregoing and following embodiments, it is also to be understood that the formulae include and represent any and all crystalline forms, partially crystalline forms, and non-crystalline and/or amorphous forms of the compounds.

The term "pharmaceutically acceptable carrier" is art-recognized and refers to a pharmaceutically-acceptable material, composition or vehicle, such as a liquid or solid filler, diluent, excipient, solvent or encapsulating material, involved in carrying or transporting any subject composition or component thereof. Each carrier must be "acceptable" in the sense of being compatible with the subject composition and its components and not injurious to the patient. Some examples of materials which may serve as pharmaceutically acceptable carriers include: (1) sugars, such as lactose, glucose and sucrose; (2) starches, such as corn starch and potato starch; (3) cellulose, and its derivatives, such as sodium carboxymethyl cellulose, ethyl cellulose and cellulose acetate; (4) powdered tragacanth; (5) malt; (6) gelatin; (7) talc; (8) excipients, such as cocoa butter and suppository waxes; (9) oils, such as peanut oil, cottonseed oil, safflower oil, sesame oil, olive oil, corn oil and soybean oil; (10) glycols, such as propylene glycol; (11) polyols, such as

glycerin, sorbitol, mannitol and polyethylene glycol; (12) esters, such as ethyl oleate and ethyl laurate; (13) agar; (14) buffering agents, such as magnesium hydroxide and aluminum hydroxide; (15) alginic acid; (16) pyrogen-free water; (17) isotonic saline; (18) Ringer's solution; (19) ethyl alcohol; (20) phosphate buffer solutions; and (21) other non-toxic compatible substances employed in pharmaceutical formulations.

As used herein, the term "administering" includes all means of introducing the compounds and compositions described herein to the patient, including, but are not limited to, oral (po), intravenous (iv), intramuscular (im), subcutaneous (sc), transdermal, inhalation, buccal, ocular, sublingual, vaginal, rectal, and the like. The compounds and compositions described herein may be administered in unit dosage forms and/or formulations containing conventional nontoxic pharmaceutically acceptable carriers, adjuvants, and vehicles.

Illustrative formats for oral administration include tablets, capsules, elixirs, syrups, and the like. Illustrative routes for parenteral administration include intravenous, intraarterial, intraperitoneal, epidural, intraurethral, infrasternal, intramuscular and subcutaneous, as well as any other art recognized route of parenteral administration.

Illustrative means of parenteral administration include needle (including microneedle) injectors, needle-free injectors and infusion techniques, as well as any other means of parenteral administration recognized in the art. Parenteral formulations are typically aqueous solutions which may contain excipients such as salts, carbohydrates and buffering agents (preferably at a pH in the range from about 3 to about 9), but, for some applications, they may be more suitably formulated as a sterile non-aqueous solution or as a dried form to be used in conjunction with a suitable vehicle such as sterile, pyrogen-free water. The preparation of parenteral formulations under sterile conditions, for example, by lyophilization, may readily be accomplished using standard pharmaceutical techniques well known to those skilled in the art. Parenteral administration of a compound is illustratively performed in the form of saline solutions or with the compound incorporated into liposomes. In cases where the compound in itself is not sufficiently soluble to be dissolved, a solubilizer such as ethanol can be applied.

The dosage of each compound of the claimed combinations depends on several factors, including: the administration method, the condition to be treated, the severity of the condition, whether the condition is to be treated or prevented, and the age, weight, and health of the person to be treated. Additionally, pharmacogenomic (the effect of genotype on the pharmacokinetic, pharmacodynamic or efficacy profile of a therapeutic) information about a particular patient may affect the dosage used.

It is to be understood that in the methods described herein, the individual components of a co-administration, or combination can be administered by any suitable means, contemporaneously, simultaneously, sequentially, separately or in a single pharmaceutical formulation. Where the co-administered compounds or compositions are administered in separate dosage forms, the number of dosages administered per day for each compound may be the same or different. The compounds or compositions may be administered via the same or different routes of administration. The compounds or compositions may be administered

9

according to simultaneous or alternating regimens, at the same or different times during the course of the therapy, concurrently in divided or single forms.

The term "therapeutically effective amount" as used herein, refers to that amount of active compound or pharmaceutical agent that elicits the biological or medicinal response in a tissue system, animal or human that is being sought by a researcher, veterinarian, medical doctor or other clinician, which includes alleviation of the symptoms of the disease or disorder being treated. In one aspect, the therapeutically effective amount is that which may treat or alleviate the disease or symptoms of the disease at a reasonable benefit/risk ratio applicable to any medical treatment. However, it is to be understood that the total daily usage of the compounds and compositions described herein may be decided by the attending physician within the scope of sound medical judgment. The specific therapeutically-effective dose level for any particular patient will depend upon a variety of factors, including the disorder being treated and the severity of the disorder; activity of the specific compound employed; the specific composition employed; the age, body weight, general health, gender and diet of the patient; the time of administration, route of administration, and rate of excretion of the specific compound employed; the duration of the treatment; drugs used in combination or coincidentally with the specific compound employed; and like factors well known to the researcher, veterinarian, medical doctor or other clinician of ordinary skill.

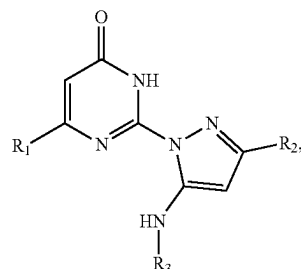
Depending upon the route of administration, a wide range of permissible dosages are contemplated herein, including doses falling in the range from about 1 µg/kg to about 1 g/kg. The dosages may be single or divided, and may administered according to a wide variety of protocols, including q.d. (once a day), b.i.d. (twice a day), t.i.d. (three times a day), or even every other day, once a week, once a month, once a quarter, and the like. In each of these cases it is understood that the therapeutically effective amounts described herein correspond to the instance of administration, or alternatively to the total daily, weekly, month, or quarterly dose, as determined by the dosing protocol.

In addition to the illustrative dosages and dosing protocols described herein, it is to be understood that an effective amount of any one or a mixture of the compounds described herein can be determined by the attending diagnostician or physician by the use of known techniques and/or by observing results obtained under analogous circumstances. In determining the effective amount or dose, a number of factors are considered by the attending diagnostician or physician, including, but not limited to the species of mammal, including human, its size, age, and general health, the specific disease or disorder involved, the degree of or involvement or the severity of the disease or disorder, the response of the individual patient, the particular compound administered, the mode of administration, the bioavailability characteristics of the preparation administered, the dose regimen selected, the use of concomitant medication, and other relevant circumstances.

The term "patient" includes human and non-human animals such as companion animals (dogs and cats and the like) and livestock animals. Livestock animals are animals raised for food production. The patient to be treated is preferably a mammal, in particular a human being.

10

In some illustrative embodiments, the invention relates to a compound of formula (I)



(I)

or a pharmaceutically acceptable salt thereof, wherein

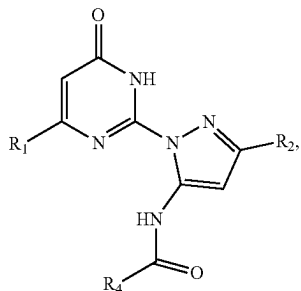
R₁ is a C₁-C₁₂ alkyl;

R₂ is a C₁-C₁₂ alkyl; and

R₃ is an acyl, alkyl, alkenyl, alkynyl, heteroalkyl, heteroalkenyl, heteroalkynyl, cycloalkyl, heterocycloalkyl, cycloalkenyl, heterocycloalkenyl, heterocyclyl, or an optionally substituted aryl, arylalkyl, or arylalkenyl.

In some embodiments, this invention relates to compounds having a general formula (I), wherein R₄ is ethyl and R₂ is methyl.

In some other embodiments, this invention relates to compounds having a general formula (II),



(II)

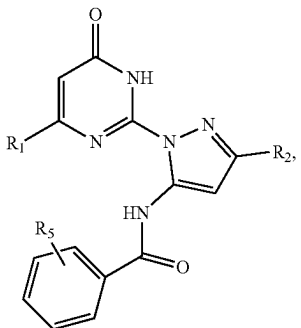
wherein

R₁ is a C₁-C₁₂ alkyl;

R₂ is a C₁-C₁₂ alkyl; and

R₄ is an alkyl, alkenyl, alkynyl, heteroalkyl, heteroalkenyl, heteroalkynyl, cycloalkyl, heterocycloalkyl, cycloalkenyl, heterocycloalkenyl, heterocyclyl, or an optionally substituted aryl, arylalkyl, or arylalkenyl.

In some embodiments, this invention relates to compounds having a general formula (III),



(III)

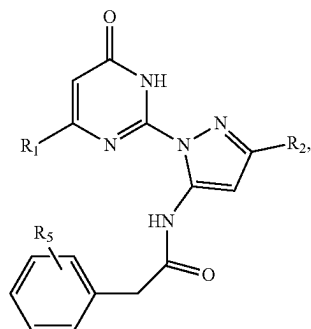
11

wherein

R₁ is a C₁-C₁₂ alkyl;R₂ is a C₁-C₁₂ alkyl; and

R₅ represents five substituents, each independently selected from the group consisting of hydrogen, deuterium, halo, azido, cyano, nitro, hydroxy, amino, thio, carboxy, ester, amide, and derivatives thereof, and acyl, sulfoxyl, sulfonyl, phosphate, phosphoryl, alkyl, alkenyl, alkynyl, heteroalkyl, heteroalkenyl, heteroalkynyl, heterocyclyl, cycloalkyl, cycloalkenyl, cycloheteroalkyl, cycloheteroalkenyl, aryl, heteroaryl, arylalkyl, arylalkenyl, and arylalkynyl, each of which is optionally substituted; or any two adjacent substituents are taken together with the attached carbons form an optionally substituted cyclic or heterocyclic moiety.

In some embodiments, this invention relates to compounds having a general formula (IV),

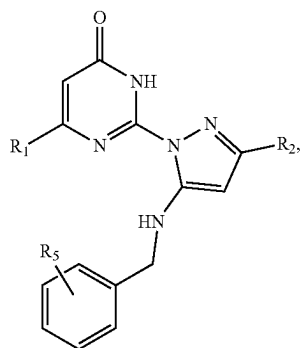


wherein

R₁ is a C₁-C₁₂ alkyl;R₂ is a C₁-C₁₂ alkyl; and

R₅ represents five substituents, each independently selected from the group consisting of hydrogen, deuterium, halo, azido, cyano, nitro, hydroxy, amino, thio, carboxy, ester, amide, and derivatives thereof, and acyl, sulfoxyl, sulfonyl, phosphate, phosphoryl, alkyl, alkenyl, alkynyl, heteroalkyl, heteroalkenyl, heteroalkynyl, heterocyclyl, cycloalkyl, cycloalkenyl, cycloheteroalkyl, cycloheteroalkenyl, aryl, heteroaryl, arylalkyl, arylalkenyl, and arylalkynyl, each of which is optionally substituted; or any two adjacent substituents are taken together with the attached carbons form an optionally substituted cyclic or heterocyclic moiety.

In some embodiments, this invention relates to compounds having a general formula (V),



12

wherein

R₁ is a C₁-C₁₂ alkyl;R₂ is a C₁-C₁₂ alkyl; and

R₅ represents five substituents, each independently selected from the group consisting of hydrogen, deuterium, halo, azido, cyano, nitro, hydroxy, amino, thio, carboxy, ester, amide, and derivatives thereof, and acyl, sulfoxyl, sulfonyl, phosphate, phosphoryl, alkyl, alkenyl, alkynyl, heteroalkyl, heteroalkenyl, heteroalkynyl, heterocyclyl, cycloalkyl, cycloalkenyl, cycloheteroalkyl, cycloheteroalkenyl, aryl, heteroaryl, arylalkyl, arylalkenyl, and arylalkynyl, each of which is optionally substituted; or any two adjacent substituents are taken together with the attached carbons form an optionally substituted cyclic or heterocyclic moiety.

In some embodiments, this invention relates to compounds selected from the group consisting of

(IV)

20

25

30

35

40

45

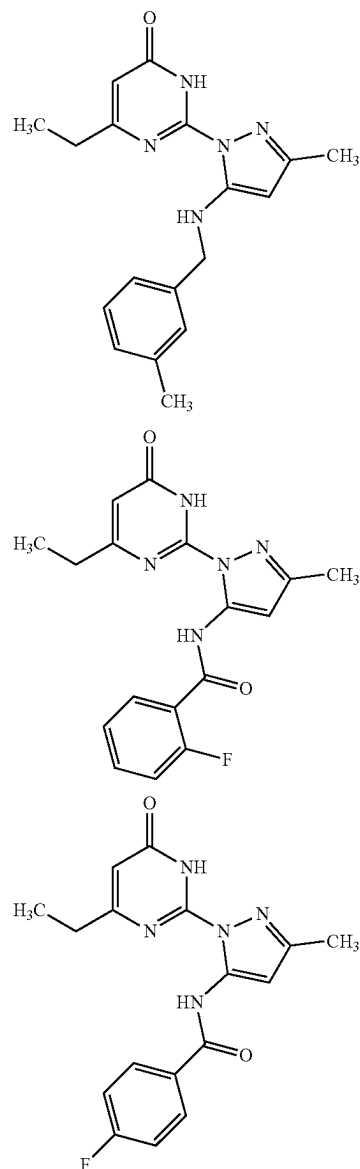
50

(V)

55

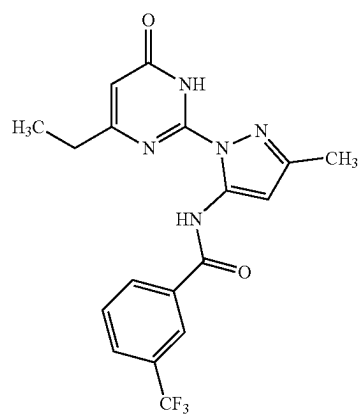
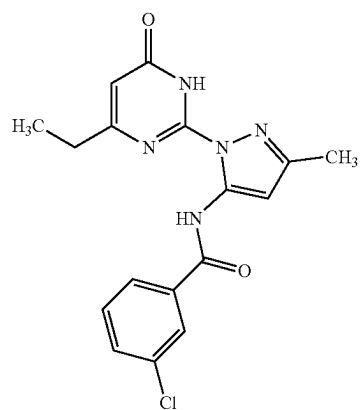
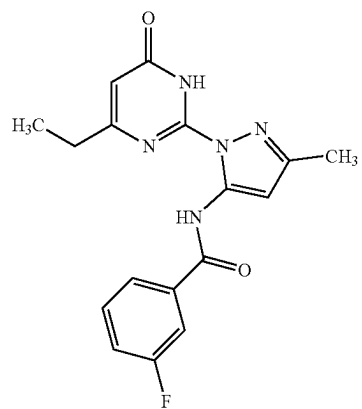
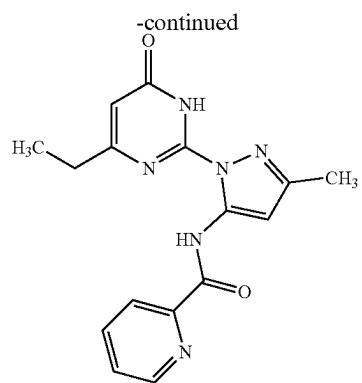
60

65

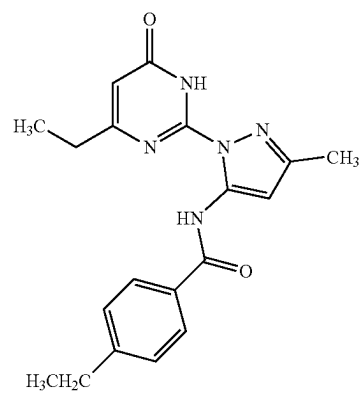
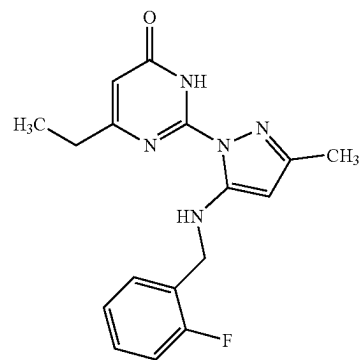
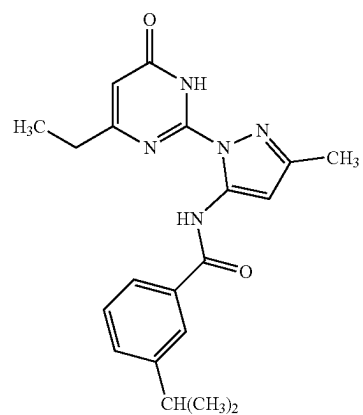
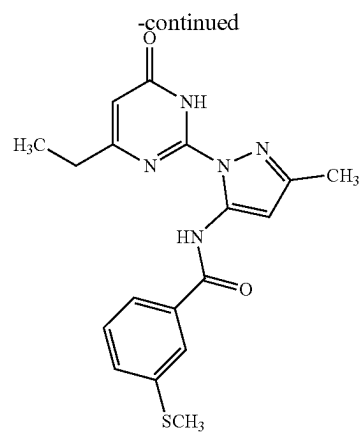


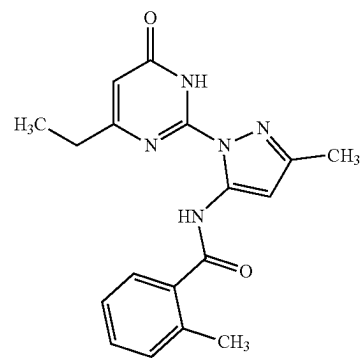
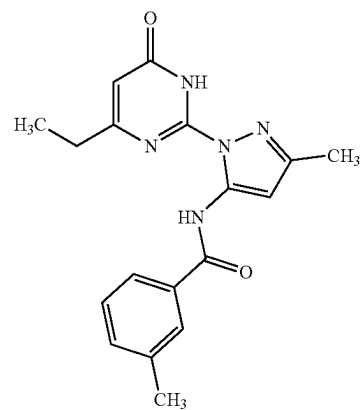
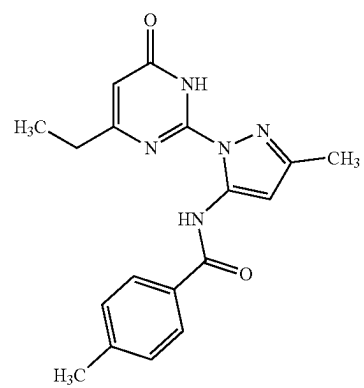
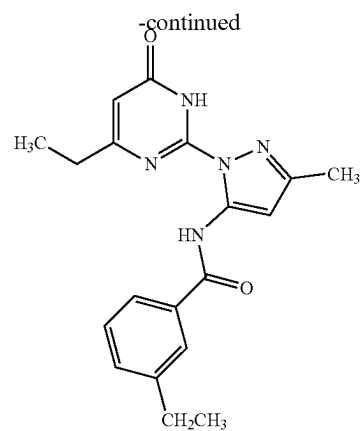
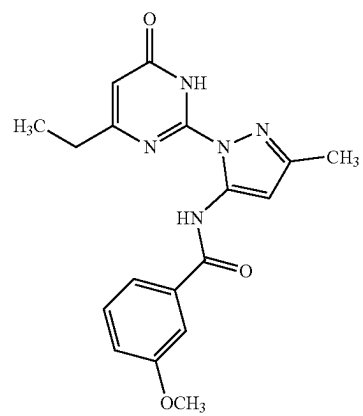
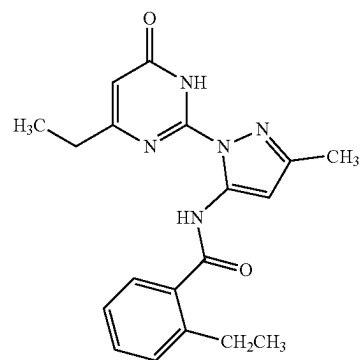
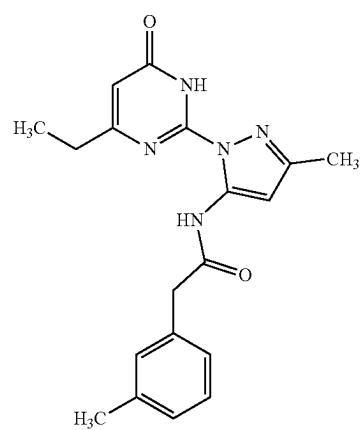
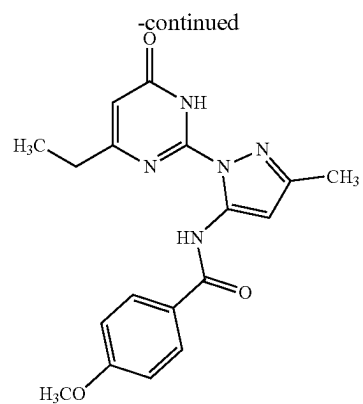
13

-continued

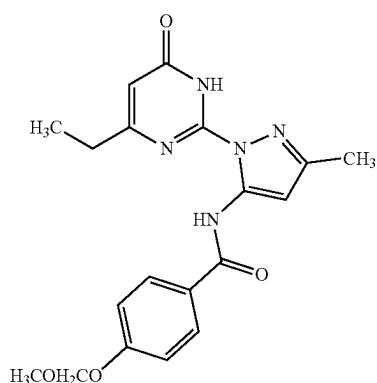
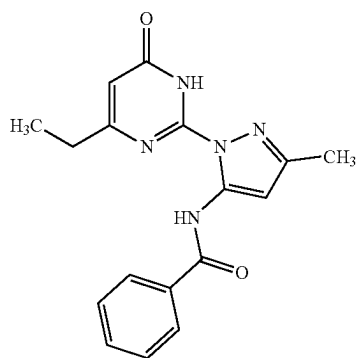
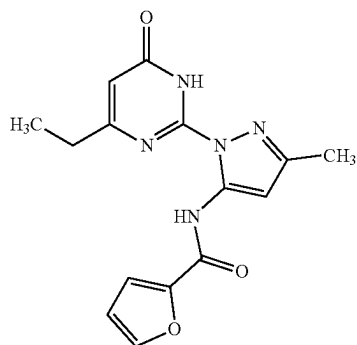
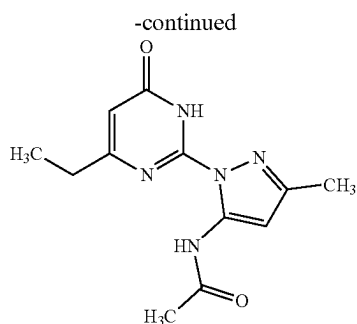
**14**

-continued



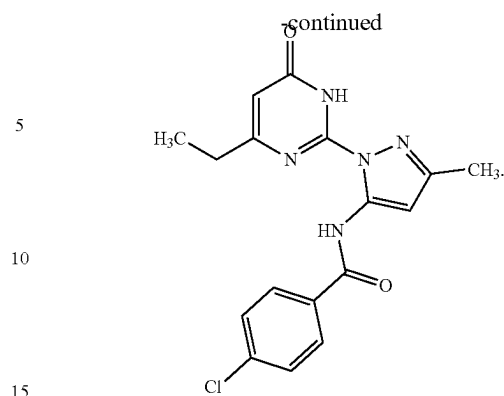
15**16**

17



and

18

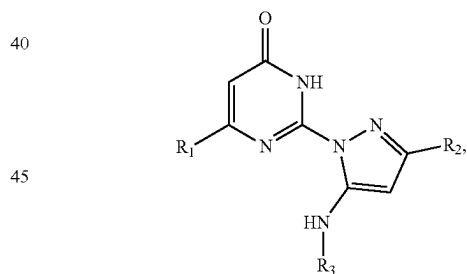


In some other embodiments, this invention relates to a pharmaceutical composition comprising one or more compounds disclosed herein, or a pharmaceutically acceptable salt thereof, together with one or more diluents, excipients or carriers.

In some other embodiments, this invention relates to a pharmaceutical composition comprising one or more compounds disclosed herein, or a pharmaceutically acceptable salt thereof, in combination with one or more other compounds by the same or different mode of action, together with one or more diluents, excipients or carriers.

In some embodiments, this invention relates to a method of treating pain, opioid dependence, alcohol use disorder, or autism comprising the step of administering to a mammal in need of relief from said pain or opioid dependence thereof a therapeutically effective amount of one or more compounds of a mammal in need thereof a therapeutically effective amount of one or more compounds of formula (I)

(I)



or a pharmaceutically acceptable salt thereof, wherein

R_1 is a C_1 - C_{12} alkyl;

R_2 is a C_1 - C_{12} alkyl; and

R_3 is an acyl, alkyl, alkenyl, alkynyl, heteroalkyl, heteroalkenyl, heteroalkynyl, cycloalkyl, heterocycloalkyl, cycloalkenyl, heterocycloalkenyl, heterocyclyl, or an optionally substituted aryl, arylalkyl, or arylalkenyl.

In some embodiments, this invention relates to method of treating pain, opioid dependence, alcohol use disorder, or autism comprising the step of administering to a mammal in need of relief from said pain or opioid dependence thereof a therapeutically effective amount of one or more compounds of a mammal in need thereof a therapeutically effective amount of one or more compounds of formula (I), wherein R_4 is ethyl and R_2 is methyl.

In some other embodiments, this invention relates to a method for treating a chronic pain.

In some other embodiments, this invention is related to a method for treating and reducing pain and opioid dependence.

In some other embodiments, this invention is related to a method for treating and reducing pain and opioid dependence further comprising the step of administering a compound of formula I in combination with an opioid drug, wherein the compound of formula I enhances μ -opioid receptor inhibition of adenylyl cyclase 1.

In some other embodiments, this invention is related to a method for treating and reducing pain and opioid dependence, wherein the opioid drug is selected from the group consisting of codeine, morphine, thebaine, oripavine, diacetylmorphine, nicomorphine, dipropanoylmorphine, diacetyldihydromorphine, acetylpropionylmorphine, desomorphine, methyl-desomorphine, dibenzoylmorphine, dihydrocodeine, ethylmorphine, heterocodeine, buprenorphine, etorphine, hydrocodone, hydromorphone, oxycodone, oxymorphone, fentanyl, alphamethylfentanyl, alfentanil, sufentanil, remifentanyl, carfentanyl, ohmefentanyl, pethidine (meperidine), ketobemidone, desmethylprodine (MPPP), allylprodine, prodine, phenethylphenylacetoxypiperidine (PEPAP), promedol, propoxyphene, dextropropoxyphene, dextromoramide, bezitramide, piritramide, methadone, dipipanone, levomethadyl acetate (LAAM), difenoxin, diphenoxylate, loperamide, dezocine, pentazocine, phenazocine, buprenorphine, dihydroetorphine, etorphine, butorphanol, nalbuphine, levorphanol, levomethorphan, lefetamine, menthol, meptazinol, mitragynine, tilidine, tramadol, tapentadol, eluxadoline, nalmefene, naloxone, and naltrexone.

The following non-limiting exemplary embodiments are included herein to further illustrate the invention. These exemplary embodiments are not intended and should not be interpreted to limit the scope of the invention in any way. It is also to be understood that numerous variations of these exemplary embodiments are contemplated herein.

Adenylyl Cyclase Inhibitors as Potential Drugs

The development and mechanistic characterization of AC1-selective small molecule inhibitors is believed to provide novel non-opioid weapons in the war on chronic pain. We identified our initial hits via screening 10,000 selected compounds from the Life Chemicals collection. The screen was carried out using HEK cells stably expressing AC1 (HEK-AC1) and cyclic AMP accumulation was stimulated using the Ca^{2+} ionophore, A23187 to selectively activate AC1. Compounds with potential chemical liabilities and PAINs were removed from further evaluation (Baell and Holloway, *J. Med. Chem.* 2010, 53, 2719-2740).

Fresh powders of several robust inhibitors (>90% inhibition) were evaluated in multiple confirmation assays assessing the dose response relationship for inhibiting AC1 and AC₈ activity. Confirmation assays used A23817 to selectively stimulate recombinant AC1 or AC8 in the HEK cell background (Cumbay and Watts (2001), *J. Pharmacol Exp Ther* 297:1201-1399). This approach takes advantage of one unique regulatory property of AC1 for the development of selective inhibitors. Both AC1 and AC8 are robustly activated by $\text{Ca}^{2+}/\text{CaM}$ under a variety of conditions, whereas the effects of $\text{Ca}^{2+}/\text{CaM}$ on AC3 are modest and conditional requiring activation by G proteins. Previous studies have identified unique $\text{Ca}^{2+}/\text{CaM}$ binding domains as one possible site of interaction to achieve AC1 (vs. AC8) selectivity (Masada et al., 2012, *Biochemistry* 51:7917-7929). These observations suggest that AC1 selectivity can be achieved through targeting $\text{Ca}^{2+}/\text{CaM}$ activation of AC1. In support of this, a portion of the hits identified for expansion herein revealed selective inhibition of $\text{Ca}^{2+}/\text{CaM}$ -stimulated AC1 versus AC8 (Table 1).

The primary technology we employed to measure cAMP accumulation was a homogenous time-resolved fluorescence (HTRF) assay, the same technology utilized in the primary screen. The activity of the new compounds was studied at AC1 and AC8 revealing a range of potency and activity patterns (Table 1).

TABLE 1

Cellular Activity Table of Compounds Disclosed

ID AC1100xx	Scaffold	R	AC1 IC50	AC8 IC50	SR
48	A	2-F	1.5	ND	40
58	A	4-Cl	0.3	9.5	32
67	A	4-OH	ND	ND	ND
68	A	none	2.7	ND	22
69	A-furan	none	6.1	ND	10
70	Phthalamide	none	ND	ND	ND
71	A	4-(OCH ₂ OCH ₃)	3.2	23	7.2
72	N-acetyl	n/a	ND	ND	ND

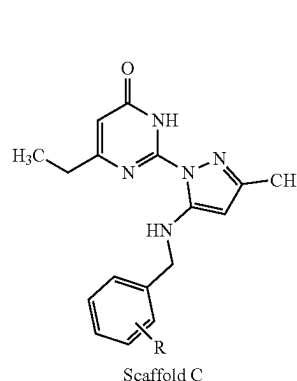
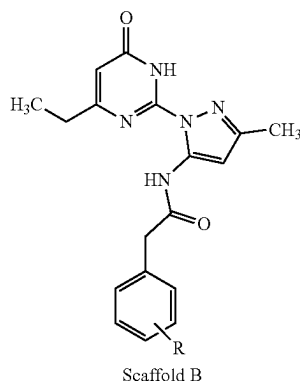
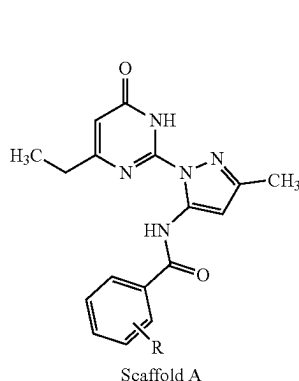


TABLE 1-continued

73	A	3-OCH ₃	1.0	ND	60
74	A	2-CH ₂ CH ₃	2.8	50	18
75	B	3-CH ₃	10	50	5.0
76	A	4-OCH ₃	0.9	33	37
77	A	2-CH ₃	2.4	ND	25
79	A	3-CH ₃	0.4	27	68
80	A	4-CH ₃	0.7	48	69
81	A	3-CH ₂ CH ₃	0.4	ND	150
82	A	4-CH ₂ CH ₃	0.5	37	74
84	C	3-CH ₃	0.2	3.3	17
85	C	2-F	0.4	5.5	14
87	A	3-CH(CH ₃) ₂	0.4	24	60
88	A	3-SCH ₃	0.4	9.7	24
89	A	3-CF ₃	0.7	18	26
90	A	3-Cl	0.4	6.9	17
91	A	3-F	1.2	12	10
92	A	4-F	0.9	14	16
93	A-2-pyridyl	none	1.2	15	13

* N/D: not determined

EXPERIMENTAL METHODS

Compounds and Other Chemicals Used

Forskolin and phorbol 12-myristate 13-acetate (PMA) were purchased from Tocris (Ellisville, MO). 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES), and ethylenediaminetetraacetic acid (EDTA) were purchased from Fisher Scientific (Pittsburg, PA). NKY80 was purchased from EMD Millipore (Temecula, CA). Isoproterenol, A23187, adenosine monophosphate (ATP), ethyleneglycol-bis(2-aminoethylether)-N,N,N',N'-tetraacetic acid (EGTA), 3-isobutyl-1-methylxanthine (IBMX), 5'-guanylyl-imidodiphosphate (GppNHp), TWEEN 20, MgCl₂, and 2-amino-2-(hydroxymethyl)-1,3-propanediol (TRIS) were purchased from Sigma-Aldrich (St. Louis, MO). 2-Bromo-1-(1-phenyl-4,5-dihydro-1H-benzo[d]azepin-3(2H)-yl)ethanone (W400), 6-chloro-2-(trichloromethyl)-4H-chromen-4-one (ST034307), and 4-chloro-1-methyl-3-nitroquinolin-2(1H)-one (ST072383) were purchased from TimTec (Newark, DE).

Cell Culture

Human embryonic kidney (HEK) cells stably expressing AC1, AC8, or AC1 with the MOR were cultured in Dulbecco's Modified Eagle Medium (Life Technologies, Grand Island, NY) supplemented with 5% bovine calf serum (Hyclone, Logan, UT), 5% fetal clone I (Hyclone), Antibiotic-Antimycotic (Life Technologies), and G418 (Invivogen, San Diego, CA) (HEK-AC1), or hygromycin B (Fisher Scientific, Pittsburg, PA) (HEK-AC8), or G418 and puromycin (Sigma-Aldrich) (HEK-AC1/MOR). Chinese hamster ovary (CHO) cells expressing the MOR (CHO-MOR) in the PathHunter® β -Arrestin GPCR assay platform were purchased from DiscoverX (Freemont, CA). Cells were grown in Ham's F12 media supplemented with 1 mM L-glutamine (Thermo Scientific, West Palm Beach, FL), 10% fetal bovine serum (Hyclone), 50 U/ml penicillin, 50 μ g/ml streptomycin (Life Technologies), G418 and hygromycin B. Cells were grown and frozen as previously described (J. M. Conley, et al., *J Vis Exp*, e51218 (2014)).

Transient Transfections

HEK cells were plated in 15 cm dishes at a confluence of 9.0×10^6 cells/dish and incubated at 37° C. in a humidified

incubator overnight. On the following day, a 6 ml solution containing 9 μ g of AC plasmid or venus fluorescent protein (venus) control plasmid, and 60 μ L lipofectamine 2000 (Life Technologies) in optiMEM (Life Technologies) was prepared and incubated at room temperature for 45 min. The solution was added dropwise to the cells, and transfection was carried out for 48 h. Cells were harvested, and cryopreserved as described above. For AC7 and AC9, HEK cells were plated in 10 cm dishes (at confluences of 3.0×10^6 or 3.5×10^6 cells/dish, respectively) and incubated at 37° C. in a humidified incubator overnight. On the following day, a 3 mL solution containing AC7 plasmid (10 μ g), AC9 plasmid (3 μ g), or venus plasmid; plus Gas plasmid (0.5 μ g for AC7 and 0.3 μ g for AC9) or venus plasmid; and Lipofectamine 2000 (48 μ L for AC7 and 24 μ L for AC9) in optiMEM was prepared and incubated at room temperature for 45 min. The solution was added dropwise to the cells, transfection was carried out for 48 h, and cells were harvested and cryopreserved.

Cyclic AMP Accumulation in Cells

Cyclic AMP accumulation was measured as previously described (Brust, et al., *J. Pharmacol. Exp. Ther.* 352, 480-493(2015)). Briefly, cryopreserved cells were thawed, resuspended in optiMEM (Life Technologies), and plated in white, flat bottom, low-volume, tissue culture-treated 384 well plates (PerkinElmer, Shelton, CT). Plates with cells were incubated in a 37° C. humidified incubator for 1 h. Inhibitors were added and plates were incubated at room temperature for 30 min followed by the addition of AC stimulants in the presence of 500 μ M IBMX. Cells were incubated at room temperature for 1 h and cAMP accumulation was measured using Cisbio's cAMP kit (Cisbio Bioassays, Bedford, MA) according to the manufacturer's instructions.

Additional assays measured cAMP accumulation using the HitHunter® cAMP Assay Platform from DiscoverX according to the manufacturer's instructions. Luminescence (HitHunter® cAMP Assay) and fluorescence (Cisbio's dynamic 2 kit) counts were measured using a Synergy 4 (BioTek, Winooski, VT).

Compound Screening

Cryopreserved HEK-AC1 cells were thawed, washed, resuspended in optiMEM and plated into white, flat bottom,

23

tissue culture-treated 384-well plates (PerkinElmer) at 15 $\mu\text{L}/\text{well}$ using a MultiFlo dispenser (BioTek). Cells were incubated in a 37° C. humidified incubator for 1 h. Next, test compounds (3.5 mg/l final assay concentration) from the NDL-3000 Natural Derivatives library (TimTec) were added (70 nL/well) using a MultiPette-mounted 384 well pin tool and incubated at room temperature for 30 min. Following the incubation with test compounds, 5 $\mu\text{L}/\text{well}$ of 3 μM A23187 in the presence of 30 nM forskolin and 500 μM IBMX (final concentrations) was added to the cells using a MultiFlo dispenser. Cells were incubated at room temperature for 1 h and cAMP accumulation was measured as described above using a MultiFlo dispenser to sequentially add 10 $\mu\text{L}/\text{well}$ of cAMP-d2 and anti-cAMP cryptate conjugate working solutions (Cisbio Bioassays) to the cells. Test compounds were screened in singlet and a Z' factor of 0.55 $\pm$ 0.22 (n=10) was obtained using 30 μM W400 as a positive control (J. M. Conley, et al., *J. Pharmacol. Exp. Ther.* 347, 276-287 (2013); J.-H. Zhang, *J. Biomol. Screen.* 4, 67-73 (1999).

Cell Viability Assays

Cell viability assays were conducted with HEK-AC1 cells following plating and compound incubation protocols identical to the procedures described above in "Cyclic AMP assays in cells". Cell viability was measured as a percentage of vehicle using 2% Triton X-100 (Sigma-Aldrich) as a control. The CellTiter-Glo® Luminescent Cell Viability Assay kit from Promega (Madison, WI) was employed to assess cell viability according to the manufacturer's instructions. Luminescence counts were measured using a Synergy 4.

Cyclic AMP Accumulation in Cellular Membranes from HEK Cells

Cellular membranes from HEK-AC1 cells were isolated and frozen as previously described in the presence of 1 mM EGTA (T. F. Brust, et al., *J. Pharmacol. Exp. Ther.* 352, 480-493 (2015)). On the assay day membranes were thawed on ice and resuspended in membrane buffer (33 mM HEPES, 0.1% TWEEN 20, 1 mM EGTA, pH 7.4). Protein concentration was measured using the Pierce BCA Protein Assay kit (Thermo Scientific) and 2.0-3.5 $\mu\text{g}/\text{well}$ was plated in a white, flat bottom, tissue culture-treated 384 well plate. Inhibitors (diluted in a 33 mM HEPES, 0.1% TWEEN 20 solution) were added and incubated for 20 min at room temperature. Next, 3 μM calmodulin or 30 μM forskolin (final concentrations) was added in stimulation buffer (33 mM HEPES, 0.1% TWEEN 20, 1.5 mM MgCl_2 , 250 μM ATP, 1 μM GppNHp, 500 μM IBMX, and 500 μM CaCl_2 —10 μM free Ca^{2+}) and incubated at room temperature for 45 min. Cyclic AMP accumulation was measured using Cisbio's dynamic 2 kit according to the manufacturer's instructions.

Adenylyl Cyclase Assays in Cellular Membranes from Sf9 Cells

Membranes from Sf9 cells expressing AC1, AC2, or AC5 were prepared as previously described (C. W. Dessauer, *Methods Enzymol.* 345, 112-126 (2002)). All activity assays were performed for 10 min at 30° C. in a final volume of 50 μL . The final concentration of MgCl_2 and Mg-ATP in the reaction was 10 mM and 200 μM , respectively. AC-containing membranes (10-20 μg) were premixed with Gas (50 nM

24

final). Inhibitors were solubilized in DMSO and incubated with AC-containing membranes for 10 min on ice before the start of the reaction. The final concentration of DMSO in the reaction did not exceed 3% for either vehicle or inhibitors. Reactions were initiated upon addition of a reaction mix containing [α - ^{32}P]ATP. The reactions were terminated with stop solution (2.5% SDS, 50 mM ATP, and 1.75 mM cAMP) and the products were separated by sequential chromatography on Dowex-50 and Al_2O_3 .

Assays with Hippocampal Homogenates

C57BL/6 mice (13 weeks old) were decapitated, their brains were quickly removed, and 2-mm slices encompassing the hippocampus were collected on ice. The hippocampal region was dissected and immediately frozen in a -80° C. freezer, where they were stored until the assay day. Dissected hippocampal tissue was thawed on ice, weighed, and homogenized in membrane buffer (2 mL/mg—wet weight) with ten manual strokes using a Wheaton-Teflon glass homogenizer. Homogenates were added to a white, flat bottom, tissue culture-treated 384-well plate and inhibitors (diluted in a 33 mM HEPES, 0.1% TWEEN 20 solution) were added and incubated for 20 min at room temperature. Next, 3 μM calmodulin (final concentration) was added in stimulation buffer (same as assays in cellular membranes from HEK cells) and incubated at room temperature for 45 min. Cyclic AMP accumulation was measured using Cisbio's dynamic 2 kit according to the manufacturer's instructions.

 β -Arrestin Recruitment Assay

Recruitment of β -arrestin 2 to the MOR was measured as previously described (T. F. Brust, et al., *J. Pharmacol. Exp. Ther.* 352, 480-493 (2015)). Briefly, CHO-MOR cells were plated in white, flat bottom, low-volume, tissue culture-treated 384-well plates. Plates with cells were incubated in a 37° C. humidified incubator overnight. Following the incubation, AC1 inhibitors or vehicle was added to the cells, which were incubated at room temperature for 30 min. Next, DAMGO or vehicle was added to cells, which were then incubated in a 37° C. humidified incubator for 1.5 h. β -arrestin 2 recruitment to the MOR was assessed using the PathHunter® assay (DiscoveRx) according to the manufacturer's instructions. Luminescence counts were measured using a Synergy 4.

Heterologous Sensitization Assays

Heterologous sensitization assays were conducted as previously described (T. F. Brust, et al., *Biochem. Pharmacol.* 93, 85-91 (2015)). Briefly, HEK-AC1/MOR cells were thawed and plated in white, flat bottom, tissue culture-treated 384-well plates. Plates with cells were incubated in a 37° C. humidified incubator for 1 h. For inhibition of the development of sensitization, inhibitors were added and plates were incubated at room temperature for 30 min, followed by addition of DAMGO and incubation at 37° C. for 2 h (to achieve sensitization). For the assays to measure inhibition of the expression of sensitization the order of DAMGO and ST034307 additions were reversed (i.e., DAMGO sensitization before AC1 inhibition). Next, cells were treated with 3 μM A23187 in the presence of 500 μM IBMX and 1 μM naloxone (final concentrations) and incubated at room temperature for 1 h. Cyclic AMP accumula-

25

tion was measured using Cisbio's dynamic 2 kit according to the manufacturer's instructions.

Animals and Housing

Wild-type C₅₇BL/6 mice, were obtained from Taconic (Cambridge City, IN). Male mice age 5 weeks (18-23 gr) were grouped and housed in single grommet ventilated plexiglass cages at ambient temperature (21° C.) in a room maintained on a reversed 12 L:12 D cycle (lights off at 10:00, lights on at 22:00) in our animal facility that has Association for Assessment and Accreditation of Laboratory Animal Care approval. Food and water were provided ad libitum. The mice were given ~7 days to acclimatize to the housing conditions and reverse light cycle before the start of the experiments. Mice were then habituated to the containment boxes for the Von Frey assay. All animal procedures were pre-approved by our Institutional Animal Care and Use Committee and were in accordance with the National Institutes of Health *Guide for the Care and Use of Laboratory Animals*. Mice were not deprived of food or water at any time.

Inflammatory Pain Behavioral Assays

C₅₇BL/6 mice were placed in suspended rectangular plastic chambers on a wire mesh grid to habituate for 1 h. Next, a baseline measurement of mechanical sensitivity to Von Frey filaments was performed as previously described (G. Corder, et al., *Science* 341, 1394-1399 (2013); R. M. van Rijn, et al., *Biol. Psychiatry* 71, 232-238 (2012)). Immediately after baseline measurements, the mice were injected with Complete Freund's adjuvant (CFA—10 μ L, non-diluted) into the intraplantar surface of the left hindpaw to induce inflammation. On the following day, inflammatory hypersensitivity was measured using Von Frey filaments. Next, drugs were injected intrathecally as previously described (T. F. Brust, et al., *Biochem. Pharmacol.* 93, 85-91

26

(2015); R. M. van Rijn, et al., *Biol. Psychiatry* 71, 232-238 (2012)). Drug-induced analgesia was measured 10 min after intrathecal injections using Von Frey filaments. Data are represented as a percentage of the average baseline response.

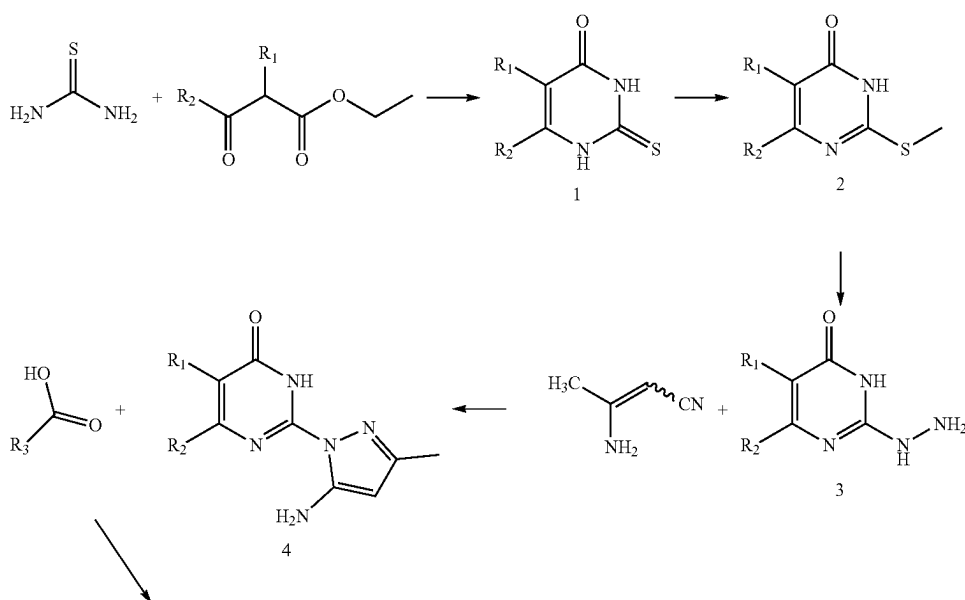
Data and Statistical Analyses

All data and statistical analyses were carried out using GraphPad Prism 6 (GraphPad Software, San Diego, CA). Statistical analyses (one-way ANOVA or t tests) are described in text or legends where appropriate.

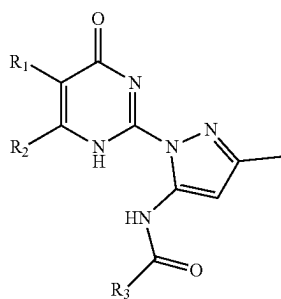
Synthetic Chemistry

The purity of all final compounds was >95% purity as assessed by HPLC according to current American Chemical Society guidelines for publication. Final compounds were analyzed on an Agilent 1200 series chromatograph. The chromatographic method utilized as Thermo Scientific Hypersil GOLD C-18 4.6 $\times$ 250 mm, 3 μ m column. UV detection wavelength=220 nm; flow-rate=1.0 mL/min; gradient=5-95% acetonitrile over 12 min and 3 min hold time at 95% acetonitrile. Both organic and aqueous mobile phases contain 0.1% v/v formic acid. The mass spectrometer used is an AB Sciex 4500 QTrap triple-quadrupole mass spectrometer with an ESI source. Samples are submitted for analysis solubilized in 1:1 acetonitrile:water solution. ¹H and ¹³C NMR spectra were recorded on either Bruker DRX500 spectrometer (operating at 500 and 125 MHz, respectively) or Bruker AVIII (operating at 800 and 200 MHz, respectively) in DMSO-d₆ or CDCl₃ with or without the internal standard of TMS at 0.05% v/v. The chemical shifts (b) reported as parts per million (ppm) and the coupling constants are reported as s=singlet, bs=broad singlet, d=doublet, t=triplet, q=quartet, dd=doublet of doublet, m=multiplet. Compounds were generally prepared according to Scheme 1 and final compounds preparation protocols are detailed below (Due-Hansen M E, et al., *Org. Biomol. Chem.* 2016, 14, 430-433).

Scheme 1. Chemical Synthesis of Pyrimidinone Series of AC1 Inhibitors
Synthetic Scheme for Pyrazolyl-pyrimidinones



-continued



5-30, with exceptions

6-ethyl-2-thioxo-2,3-dihydropyrimidin-4(1H)-one
(1)

To a 250 mL round bottom flask equipped with stirbar, thiourea (5.9895 g, 78.68 mmol, 1 eq.) and potassium hydroxide (5.0996 g, 90.88 mmol, 1.155 eq.) are added. The powders are then suspended with 60 mL of 200 proof ethanol and the mixture is heated while stirring to 75° C. To this stirring mixture, ethyl 3-oxopentanoate (12.5 mL, 87.6 mmol, 1.11 eq.) is added in one portion. The flask is then sealed with a glass stopper and the solution is heated at 85° C. overnight. White solid began crashing out of solution quickly, with the solution appearing orange/pink.

The following morning (~16 hours), the suspension is removed from heat and is allowed to slowly return to room temperature. The suspension is concentrated on rotavap to remove about half of the volume (~30 mL ethanol remaining). To this suspension, deionized water is added slowly while stirring until the mixture turns completely clear. The solution is then slowly neutralized while stirring by adding concentrated HCl (12 N in water) and diluted NaOH (1.0 N in water). pH is ascertained with litmus paper. As the solution approaches neutral pH, white solid should begin crashing out of solution (if the solution becomes too thick to stir, additional water is added as needed). Once the mixture is neutralized, the suspension is then filtered and washed several times with water. If diffracting crystals are seen (remaining thiourea), wash with several mL of ethanol to remove contaminant. The white powder is then allowed to dry on filter paper for at least 1-2 hours before being transferred to a vial and placed on hi-vac. The resultant powder is collected as product without further purification (7.4626 g, 78.68 mmol, 60.72% yield). ¹H NMR (500 MHz, Methanol-d₄) δ 5.70 (s, 1H), 2.43 (q, J=7.5 Hz, 2H), 1.21 (t, J=7.5 Hz, 3H).

6-ethyl-2-(methylthio)pyrimidin-4(1H)-one (2)

Compound (1) (7.4626 g, 78.68 mmol, 1 eq.) is added to a 250 mL round bottom flask equipped with stirbar. In a separate container, sodium hydroxide (2.102 g, 52.554 mmol, 1.1 eq.) is added and dissolved via sonication in 30 mL of water. This solution is then added slowly while stirring to the round bottom flask. The solution is allowed to stir for 30 minutes at room temperature. The solution is then placed in an ice water bath (0° C.) for 5 minutes (note: some solid may fall out of solution). To this stirring mixture in the ice bath, iodomethane (3.57 mL, 57.331 mmol, 1.2 eq.) is added in one portion. The mixture is stirred rapidly to ensure distribution of iodomethane, and the mixture is allowed to stir overnight inside the ice water bath (0-23° C.).

The following morning the suspension is immediately filtered, washed with several mL of ice-cold water, and allowed to dry for 15 minutes on filter paper before transfer to hi-vac. The powder is then collected, yielding a white powder as product without further purification (7.4606 g, 47.776 mmol, 91.734%). ¹H NMR (500 MHz, DMSO-d₆) δ 12.50 (s, 1H), 5.90 (s, 1H), 2.44 (s, 3H), 2.39 (q, J=7.4 Hz, 2H), 1.09 (t, J=7.5 Hz, 3H).

6-ethyl-2-hydrazineylidene-2,3-dihydropyrimidin-4(1H)-one (3)

To a 250 mL round bottom flask equipped with stirbar, (7.4606 g, 43.827 mmol, 1 eq.), potassium carbonate (61.2 mg, 443 μmol, 1 mol %), and 30 mL of 2-propanol are added. To this stirring mixture, hydrazine hydrate (10.6 mL, 219 mmol, 4.99 eq., 64% hydrazine) is added in one portion. The vessel is tightly sealed with a glass stopper and heated to 80° C. for 5 hours and then lowered to 60° C. overnight.

The following morning, the reaction is cooled briefly on ice and the precipitate is filtered and washed several times with 1 mL volumes of diethyl ether and methanol. The resultant powder is an off-white, refractive powder. It was then dried on hi-vac for several hours. The powder is taken forward without further purification (2.7798 g, 18.031 mmol, 41.141%). ¹H NMR (500 MHz, DMSO-d₆) δ 8.45 (s, 1H), 5.36 (s, 1H), 4.51 (s, 2H), 2.27 (q, J=7.5 Hz, 2H), 1.08 (t, J=7.5 Hz, 3H).

2-(5-amino-3-methyl-1H-pyrazol-1-yl)-6-ethylpyrimidin-4(1H)-one (4)

(3) (2.7798 g, 18.031 mmol, 1 eq.) and 3-aminocrotonitrile (2.9610 g, 36.06 mmol, 2 eq.) are added to a 250 mL round bottom flask equipped with stirbar and then suspended in 30 mL of 200 proof ethanol. The vessel is sealed with a rubber stopper and vented with a 20 G needle. The mixture is heated at 90° C. for 4 hours then 60° C. overnight.

The following morning, the reaction is unsealed and allowed to return to room temperature while stirring. The ethanolic mixture is then re-sealed and chilled to 3° C. inside a refrigerator for five minutes. The precipitate is then filtered and washed with several 10 mL portions of hexanes. The precipitate is then washed with a 10 mL portion of ethanol, 1 mL of methanol dropwise, and 1 mL of DCM dropwise. The resultant powder was then allowed to dry on filter for 15 minutes before transfer to hi-vac. The resultant off-white/tan powder typically required no further purification (2.4926 g, 11.081 mmol, 61.458%).

For smaller batches, the powder could be further purified if required via normal phase flash chromatography (DCM:

29

MeOH 10% isocratic). Fractions containing product were identified via UV absorbance at 254 nm and corroborated via APCI-MS: m/z 220.1 $[M+H]^+$. 1H NMR (500 MHz, DMSO- d_6) δ 11.47 (s, 1H), 6.90 (s, 2H), 5.98 (s, 1H), 5.27 (s, 1H), 2.52 (q, $J=7.6$ Hz, 2H), 2.09 (s, 3H), 1.16 (t, $J=7.5$ Hz, 3H).

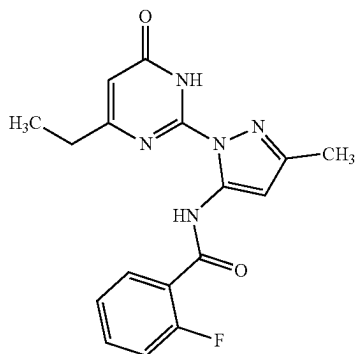
General Procedure A for Amide Coupling with Benzoic Acids and Derivatives (5-30*)

Acyl fluoride amide coupling is based off a previously reported procedure¹. Carboxylic acids (primarily benzoic acids) (0.296 mmol, 1.3 eq.) and Fluoro- N,N,N',N' -bis(tetramethylene)formamidine hexafluorophosphate (BTFFH) (0.342 mmol, 1.5 eq.) are added to a 0.5-2 mL thick-walled Biotage microwave vial with magnetic stirbar. The vial is sealed and flushed with argon. To this vial, dry dichloromethane (DCM) (440 μ L) is added. Next, N,N -diisopropylethylamine (DIPEA) (180 μ L, 1.03 mmol, 4.5 eq.) is added to the vial, and the mixture is allowed to stir for one hour at room temperature under an Argon balloon.

(4) (0.228 mmol, 1 eq.) is then added to a separate 0.5-2 mL Biotage microwave vial with magnetic stirbar. The vial is sealed and then flushed with argon. Contents of the first vial are then added dropwise to the amine. This vial is then heated at 90° C. in sand while stirring for 12-24 hours.

The reaction is then quenched with 1 mL 10 mM potassium phosphate buffer (pH 7), agitated, and then allowed to stir for at least 1 hour at room temperature. The vial contents are then added to a separatory flask and diluted with 10 mL 10 mM potassium phosphate buffer. The aqueous layer is then extracted twice with 10 mL dichloromethane. The combined organic layers are then washed twice with 20 mL 0.12 N HCl and once with 20 mL brine. The organic layer is then dried with $MgSO_4$, filtered, and concentrated on rotavap, yielding an off-white film. The resultant film was then suspended with diethyl ether and concentrated on rotavap to remove remaining DCM, yielding a powder. This powder was then suspended in 2 mL ethyl acetate, filtered, and allowed to dry, yielding product as a pure powder without further purification.

N -(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)-2-fluorobenzamide (5)
AC10048

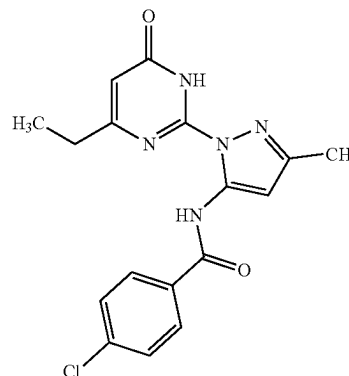


Prepared using general procedure A with 2-fluorobenzoic acid (83.1 mg, 593 μ mol), BTFFH 216.0 mg, 684 μ mol), DCM (880 μ L), DIPEA (360 μ L, 2.1 mmol) and (4) (100.0 mg, 456.1 μ mol) to produce AC10048 (59.3 mg, 174 μ mol, 38.1%) as a white solid. 1H NMR (800 MHz, $CDCl_3$): δ

30

12.36 (s, 1H), 10.31 (s, 1H), 8.15 (td, $J=7.8, 1.8$ Hz, 1H), 7.60-7.55 (m, 1H), 7.34 (td, $J=7.5, 1.1$ Hz, 1H), 7.22 (ddd, $J=11.9, 8.3, 1.0$ Hz, 1H), 6.97 (s, 1H), 6.09 (s, 1H), 2.63 (q, $J=7.5$ Hz, 2H), 2.31 (s, 3H), 1.24 (t, $J=7.5$ Hz, 3H) and ^{13}C NMR (126 MHz, Chloroform- d) δ 169.04, 161.54-159.22 (m), 153.75, 148.27, 140.38, 134.49, 134.41, 132.35, 125.14 (d, $J=3.5$ Hz), 120.63 (d, $J=11.6$ Hz), 116.32, 116.14, 107.65, 100.22, 30.09, 14.17, 12.16. APCI-MS(+): m/z 342.2 $[M+H]^+$, 202.0. APCI-MS(-): m/z 340.0 $[M-H]^-$. HPLC retention time: 12.654 min. HPLC purity 98.7%.

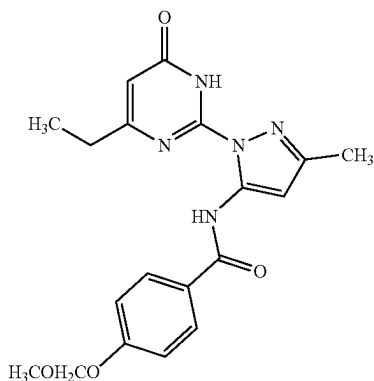
4-chloro- N -(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)benzamide (6)
AC10058



Prepared using general procedure A with 4-chlorobenzoic acid (46.4 mg, 297 μ mol), BTFFH (108.0 mg, 341.6 μ mol), DCM (440 μ L), DIPEA (180 μ L, 1.0 mmol), and (4) (50.1 mg, 229 μ mol) to produce AC10058 (26.8 mg, 74.9 μ mol, 32.8%) as a white solid. 1 standard parameters, cryoprobe. 1H NMR (800 MHz, Chloroform- d) δ 12.40 (s, 1H), 10.32 (s, 1H), 7.94-7.93 (m, 2H), 7.54-7.52 (m, 2H), 6.89 (s, 1H), 6.13 (s, 1H), 2.66 (q, $J=7.5$ Hz, 2H), 1.32 (t, $J=7.5$ Hz, 3H). 1H NMR (800 MHz, Chloroform- d) δ 12.38 (s, 1H), 10.30 (s, 1H), 7.93-7.91 (m, 2H), 7.52-7.50 (m, 2H), 6.87 (s, 1H), 6.11 (s, 1H), 2.64 (q, $J=7.5$ Hz, 2H), 2.31 (s, 3H), 1.30 (t, $J=7.5$ Hz, 3H). ^{13}C NMR (201 MHz, Chloroform- d) δ 167.80, 162.42, 160.80, 154.18, 148.77, 140.89, 139.17, 131.63, 129.23, 128.68, 107.79, 99.13, 30.66, 14.23, 12.27. APCI-MS(+): m/z 360.1, 358.1 $[M+H]^+$, 220.0, 218.0, 141.0, 139.0. APCI-MS(-): m/z 358.0, 355.9 $[M-H]^-$. HPLC retention time: 13.357 min. HPLC purity >99.0%.

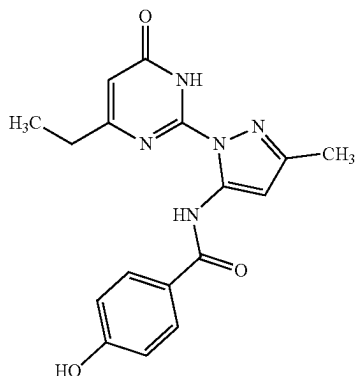
31

N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)-4-(methoxymethoxy)benzamide (7) AC10071



Prepared using general procedure A with 4-(methoxymethoxy)benzoic acid (54.0 mg, 296 μ mol), BTFFH (108.0 mg, 341.6 μ mol), DCM (440 μ L), DIPEA (180 μ L, 1.0 mmol), and (4) (50.0 mg, 228 μ mol) to produce AC10071 (14.9 mg, 38.9 μ mol, 17.0%) as a white solid. ^1H NMR (800 MHz, Chloroform- d) δ 12.27 (s, 1H), 7.94 (d, $J=8.3$ Hz, 2H), 7.14 (d, $J=8.5$ Hz, 2H), 6.87 (s, 1H), 6.11 (s, 1H), 3.51 (s, 2H), 2.66 (q, $J=7.5$ Hz, 2H), 2.30 (s, 3H), 1.31 (t, $J=7.5$ Hz, 3H). ^{13}C NMR (201 MHz, Chloroform- d) δ 168.05, 163.01, 160.83, 154.18, 148.78, 141.30, 129.21, 126.45, 116.14, 107.63, 98.80, 94.20, 56.35, 30.70, 29.73, 14.19, 12.33. APCI-MS(+): m/z 384.1 [M+H]. APCI-MS(-): m/z 382.0 [M-H]. HPLC retention time: XX.XXX min. HPLC purity >95%

N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)-4-hydroxybenzamide (8) AC10067

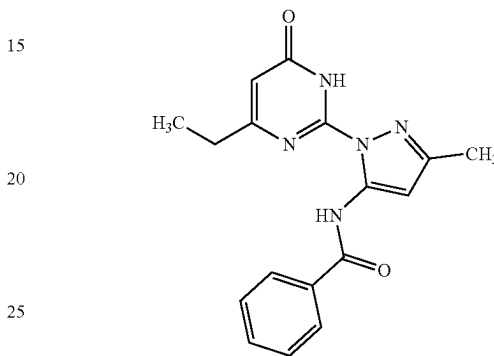


(7) (14.9 mg, 38.9 μ mol) was dissolved in DCM (5 mL) and allowed to cool to 0 $^{\circ}\text{C}$ in an ice water bath. To this solution, excess trifluoroacetic acid (TFA) was added. It was allowed to stir for 3 hours at 0 $^{\circ}\text{C}$. The solution was concentrated on rotavap, resuspended with hexanes, then filtered, yielding AC10067 (5.5 mg, 16 μ mol, 42%) as a white solid. ^1H NMR (800 MHz, DMSO- d_6) δ 12.53 (s, 1H), 10.35 (s, 1H), 7.82 (d, $J=8.2$ Hz, 2H), 6.93 (d, $J=8.2$ Hz,

32

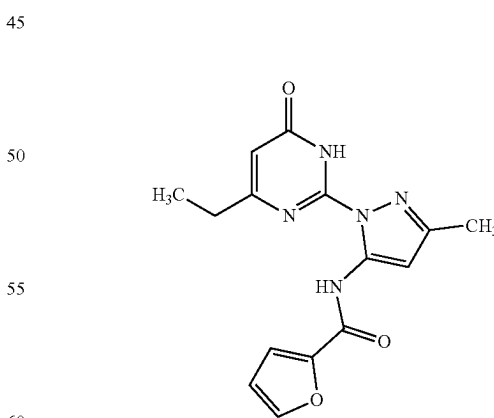
2H), 6.75 (s, 1H), 6.29 (s, 1H), 2.68 (t, $J=7.9$ Hz, 2H), 2.26 (s, 3H), 1.21 (t, $J=7.7$ Hz, 3H). ^{13}C NMR (201 MHz, DMSO- d_6) δ 162.83, 161.96, 151.90, 141.33, 129.71, 124.19, 116.03, 97.67, 30.24, 14.35, 12.95. APCI-MS(+): m/z 340.2 [M+H]. APCI-MS(-): m/z 452.0 [M+TFA-H], 338.0 [M-H]. HPLC retention time: 10.822 min. HPLC purity >99.0%.

N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)benzamide (9) AC10068



Prepared using general procedure A with benzoic acid (36.2 mg, 296 μ mol), BTFFH (108.0 mg, 341.6 μ mol), DCM (440 μ L), DIPEA (180 μ L, 1.0 mmol), and (4) (50.0 mg, 228 μ mol) to produce AC10068 (16.7 mg, 51.6 μ mol, 22.6%) as a white solid. ^1H NMR (500 MHz, Chloroform- d) δ 12.38 (s, 1H), 10.41-10.25 (m, 1H), 8.00-7.95 (m, 2H), 7.64-7.59 (m, 1H), 7.53 (dd, $J=8.4$, 7.0 Hz, 2H), 6.88 (s, 1H), 6.10 (s, 1H), 2.65 (q, $J=7.5$ Hz, 2H), 2.31 (s, 3H), 1.29 (t, $J=7.5$ Hz, 3H). ^{13}C NMR (126 MHz, Chloroform- d) δ 167.87, 163.37, 160.79, 154.02, 148.62, 140.98, 133.07, 132.61, 128.79, 127.17, 107.58, 98.85, 30.52, 14.13, 12.17. APCI-MS(+): m/z 324.1 [M+H]. APCI-MS(-): m/z 322.0 [M-H].

N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)furan-2-carboxamide (10) AC10069



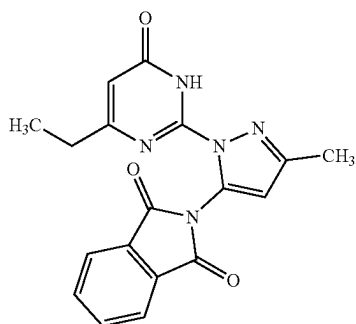
Furan-2-carboxylic acid (31.0 mg, 0.27 mmol, 3 eq.) and DIPEA (50 mg, 67 μ L, 4.2 eq., 0.38 mmol) were dissolved completely in DMF (0.9 mL) and allowed to stir at room temperature for 10 minutes. Next, 2-(3H-[1,2,3]triazolo[4,5-b]pyridin-3-yl)-1,1,3,3-tetramethylisouronium hexafluorophosphate(V) (0.11 g, 3.2 Eq, 0.29 mmol) was added to

33

the vial. This solution was allowed to stir for 1 hour at room temperature. The color changed from clear to dark violet/orange/maroon.

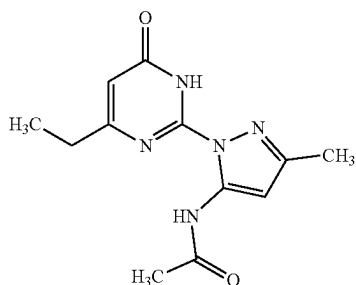
To this stirring solution, (4) (20 mg, 1 Eq, 91 μ mol) was added. This solution was then brought to 85 C and allowed to stir overnight. Work-up was the same as general procedure A, yielding AC10069 as a white solid (2.9 mg, 9.3 μ mol, 10%). ^1H NMR (800 MHz, Chloroform- d) δ 12.49 (s, 1H), 10.25 (s, 1H), 7.53 (d, $J=1.7$ Hz, 1H), 7.31 (d, $J=3.5$ Hz, 1H), 6.83 (s, 1H), 6.62 (dd, $J=3.5, 1.7$ Hz, 1H), 6.10 (s, 1H), 2.69 (q, $J=7.6$ Hz, 2H), 2.30 (s, 3H), 1.37 (t, $J=7.6$ Hz, 3H). ^{13}C NMR (201 MHz, Chloroform- d) δ 168.39, 161.04, 154.33, 153.99, 148.53, 147.22, 144.65, 140.34, 116.41, 113.00, 107.77, 98.99, 30.59, 14.21, 12.17.

2-(1-(4-ethyl-6-hydroxypyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)isoindoline-1,3-dione (11)
AC10070



Prepared using general procedure A with 2-(methoxycarbonyl)benzoic acid (53.4 mg, 296 μ mol), BTFFH (108.0 mg, 341.6 μ mol), DCM (440 μ L), DIPEA (180 μ L, 1.0 mmol), and (4) (50.0 mg, 228 μ mol) to produce AC10070 (4.0 mg, 11 μ mol, 5.0%) as a white solid. ^1H NMR (800 MHz, Chloroform- d) δ 10.20 (s, 1H), 7.99 (dd, $J=5.5, 3.0$ Hz, 2H), 7.86 (dd, $J=5.5, 3.0$ Hz, 2H), 6.52 (s, 1H), 5.94 (s, 1H), 2.40 (s, 3H), 1.77 (q, $J=7.4$ Hz, 2H), 0.67 (t, $J=7.5$ Hz, 3H). ^{13}C NMR (201 MHz, Chloroform- d) δ 169.24, 166.46, 161.24, 152.33, 145.89, 134.78, 132.39, 131.22, 124.17, 113.00, 108.48, 30.30, 14.32, 11.64. APCI-MS(+): m/z 350.2 [M+H]. APCI-MS(-): m/z 348.0 [M-H]. HPLC retention time: 11.296 min. HPLC purity 96.02%.

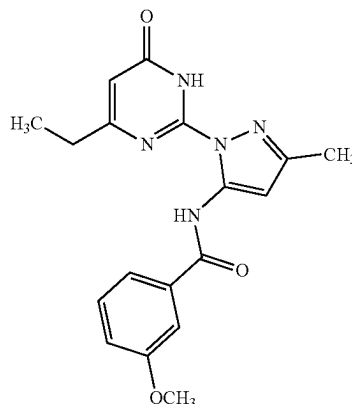
N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)acetamide (12) AC10072



34

(4) (25.0 mg, 114 μ mol) was suspended in 2-MeTHF (500 μ L) inside a Biotage microwave vial. Acetylsalicylic acid (30.8 mg, 171 μ mol) and pyridine (30 μ L, 370 μ mol) were added to the stirring mixture. Finally, propylphosphonic anhydride (50% w/w in EtOAc) (145 mg, 228 μ mol) was added to the mixture which was then immediately sealed. The vial was then heated to 100 C in a Biotage Initiator microwave for 12 hours. After cooling, the mixture was diluted with DI H₂O and agitated vigorously. The aqueous layer was then extracted with an equal volume of DCM. The organic layer was then collected, dried with MgSO₄, and concentrated on rotavap. It was then further purified via normal phase flash chromatography (2-6% MeOH:DCM). Fractions containing product identified by APCI MS and UV absorbance at 254 nm were concentrated and dried on hi-vac yielding AC10072 (10.2 mg, 39.0 μ mol, 34.2%) as an off-white solid. ^1H NMR (800 MHz, Chloroform- d) δ 11.53 (s, 1H), 6.72 (s, 1H), 6.07 (s, 1H), 2.61 (q, $J=7.6$ Hz, 2H), 2.27 (s, 3H), 2.24 (s, 3H), 1.30 (td, $J=7.5, 1.6$ Hz, 3H). ^{13}C NMR (201 MHz, Chloroform- d) δ 167.84, 166.34, 160.90, 153.96, 148.60, 140.68, 107.77, 98.92, 30.28, 24.33, 14.16, 11.89. APCI-MS(+): m/z 262.0 [M+H]. APCI-MS(-): m/z 259.9 [M-H]. HPLC retention time: 10.065 min. HPLC purity 98.94%.

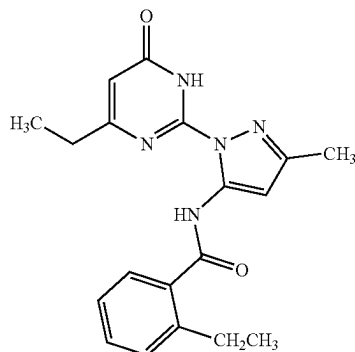
N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)-3-methoxybenzamide (13)
AC10073



Prepared using general procedure A with 3-methoxybenzoic acid (45.1 mg, 296 μ mol), BTFFH (108.0 mg, 341.6 μ mol), DCM (440 μ L), DIPEA (180 μ L, 1.0 mmol), and (4) (50.0 mg, 228 μ mol) to produce AC10073 (27.5 mg, 77.8 μ mol, 34.1%) as a white solid. ^1H NMR (800 MHz, Chloroform- d) δ 12.34 (s, 1H), 10.33 (s, 1H), 7.54 (s, 1H), 7.49 (d, $J=7.7$ Hz, 1H), 7.41 (t, $J=7.9$ Hz, 1H), 7.13 (d, $J=8.3$ Hz, 1H), 6.86 (s, 1H), 6.09 (s, 1H), 3.88 (s, 3H), 2.65 (q, $J=7.6$ Hz, 2H), 2.30 (s, 3H), 1.28 (t, $J=7.6$ Hz, 3H). ^{13}C NMR (201 MHz, Chloroform- d) δ 168.12, 163.31, 160.93, 160.22, 154.08, 148.71, 141.06, 134.61, 129.78, 118.79, 118.32, 113.37, 107.66, 98.94, 55.57, 30.63, 14.22, 12.29. APCI-MS(+): m/z 354.1 [M+H], 299.1. APCI-MS(-): m/z 352.0 [M-H]. HPLC retention time: 12.600 min. HPLC purity >99.0%.

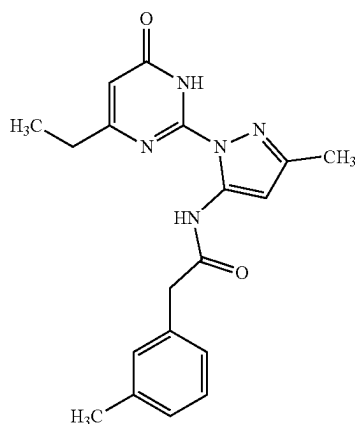
35

2-ethyl-N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)benzamide (14)
AC10074



Prepared using general procedure A with 2-ethylbenzoic acid (44.5 mg, 296 μ mol), BTFFH (108.0 mg, 341.6 μ mol), DCM (440 μ L), DIPEA (180 μ L, 1.0 mmol), and (4) (50.0 mg, 228 μ mol) to produce AC10074 (4.2 mg, 12 μ mol, 5.2%) as a white solid. ^1H NMR (800 MHz, Chloroform- d) δ 11.99 (s, 1H), 10.28 (s, 1H), 7.58 (dd, $J=7.6$, 1.4 Hz, 1H), 7.46 (td, $J=7.6$, 1.3 Hz, 1H), 7.37-7.34 (m, 1H), 7.28 (td, $J=7.5$, 1.2 Hz, 1H), 6.89 (s, 1H), 6.04 (d, $J=1.0$ Hz, 1H), 2.93 (q, $J=7.5$ Hz, 2H), 2.43 (q, $J=7.7$ Hz, 2H), 2.31 (s, 3H), 1.27 (t, $J=7.5$ Hz, 3H), 1.06 (t, $J=7.6$ Hz, 3H). ^{13}C NMR (201 MHz, Chloroform- d) δ 168.10, 165.82, 160.96, 154.08, 148.64, 144.23, 140.99, 134.13, 131.34, 130.32, 126.82, 125.86, 107.67, 98.81, 30.27, 26.62, 16.01, 14.20, 11.83. APCI-MS(+): m/z 352.2 [M+H], 220.1, 133.1. APCI-MS(-): m/z 350.1 [M-H]. HPLC retention time: 13.365 min. HPLC purity 97.33%.

N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)-2-(*m*-tolyl)acetamide (15)
AC10075

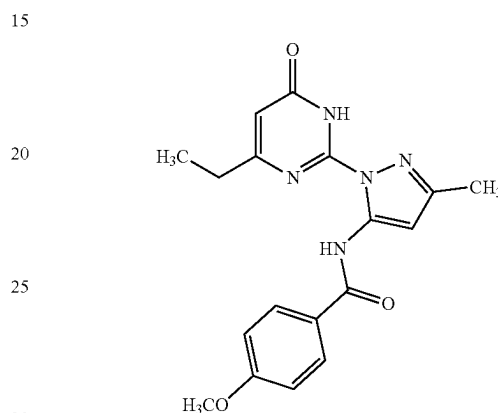


Prepared using general procedure A with 2-(*m*-tolyl) acetic acid (44.6 mg, 297 μ mol), BTFFH (108.0 mg, 341.6 μ mol), DCM (440 μ L), DIPEA (180 μ L, 1.0 mmol), and (4) (50.1 mg, 229 μ mol) to produce AC10075 (17.8 mg, 50.7 μ mol, 22.2%) as a white solid. ^1H NMR (800 MHz, Chloroform- d) δ 11.35 (s, 1H), 10.19 (s, 1H), 7.25 (t, $J=7.5$ Hz,

36

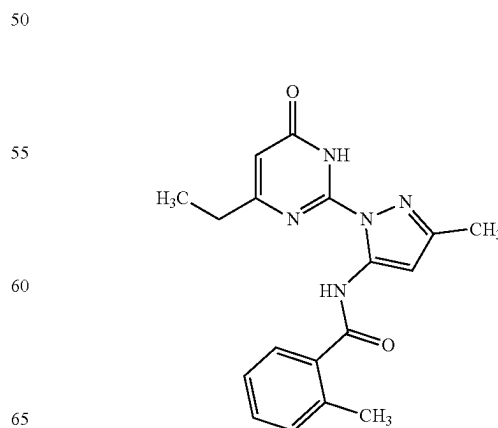
1H), 7.14-7.11 (m, 3H), 6.77 (s, 1H), 6.00 (s, 1H), 3.75 (s, 2H), 2.35 (s, 3H), 2.30 (q, $J=7.5$ Hz, 2H), 2.25 (s, 3H), 1.19 (t, $J=7.5$ Hz, 3H). ^{13}C NMR (201 MHz, Chloroform- d) δ 168.25, 167.94, 160.95, 153.81, 148.36, 140.60, 138.82, 133.51, 130.01, 129.00, 128.45, 126.27, 107.40, 99.21, 44.86, 30.20, 21.38, 14.13, 12.09. APCI-MS(+): m/z 352.2 [M+H], 260.2. APCI-MS(-): m/z 350.1 [M-H]. HPLC retention time: 12.694 min. HPLC purity 96.60%.

N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)-4-methoxybenzamide (16)
AC10076



Prepared using general procedure A with 4-methoxybenzoic acid (45.1 mg, 296 μ mol), BTFFH (108.0 mg, 341.6 μ mol), DCM (440 μ L), DIPEA (180 μ L, 1.0 mmol), and (4) (50.0 mg, 228 μ mol) to produce AC10076 (29.2 mg, 82.6 μ mol, 36.2%) as a white solid. ^1H NMR (800 MHz, Chloroform- d) δ 12.25 (s, 1H), 10.31 (s, 1H), 7.96-7.93 (m, 2H), 7.02-6.98 (m, 2H), 6.86 (s, 1H), 6.10 (d, $J=1.1$ Hz, 1H), 3.90 (s, 3H), 2.66 (q, $J=7.5$ Hz, 2H), 2.30 (s, 3H), 1.31 (t, $J=7.6$ Hz, 3H). ^{13}C NMR (201 MHz, Chloroform- d) δ 168.01, 163.20, 163.10, 160.94, 154.19, 148.78, 141.37, 129.27, 125.47, 114.11, 107.63, 98.73, 55.58, 30.69, 14.23, 12.32. APCI-MS(+): m/z 354.2 [M+H], 135.0. APCI-MS(-): m/z 352.1 [M-H]. HPLC retention time: 12.451 min. HPLC purity 98.78%.

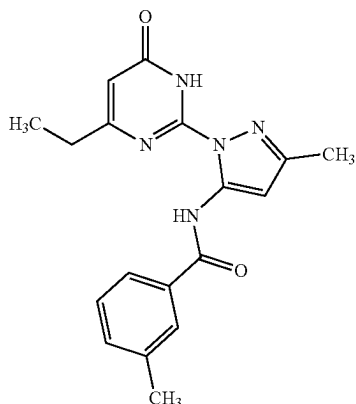
N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)-2-methylbenzamide (17)
AC10077



37

Prepared using general procedure A with 2-methylbenzoic acid (40.4 mg, 296 μ mol), BTFFH (108.0 mg, 341.6 μ mol), DCM (440 μ L), DIPEA (180 μ L, 1.0 mmol), and (4) (50.0 mg, 228 μ mol) to produce AC10077 (5.6 mg, 17 μ mol, 7.3%) as a white solid. ¹H NMR (800 MHz, Chloroform-d) δ 12.01 (s, 1H), 10.27 (s, 1H), 7.63-7.61 (m, 1H), 7.42 (td, J=7.5, 1.4 Hz, 1H), 7.32 (d, J=7.6 Hz, 1H), 7.29 (t, J=7.5 Hz, 1H), 6.89 (s, 1H), 6.05 (d, J=0.9 Hz, 1H), 2.58 (s, 3H), 2.46 (q, J=7.5 Hz, 2H), 2.31 (s, 3H), 1.10 (t, J=7.6 Hz, 3H). ¹³C NMR (201 MHz, Chloroform-d) δ 168.09, 165.65, 160.93, 154.09, 148.65, 141.01, 138.19, 134.22, 131.94, 131.29, 126.85, 125.88, 107.68, 98.79, 30.32, 20.37, 14.21, 11.92. APCI-MS(+): m/z 338.2 [M+H], 220.0, 119.0. APCI-MS(-): m/z 336.1 [M-H]. HPLC retention time: 12.897 min. HPLC purity 97.51%.

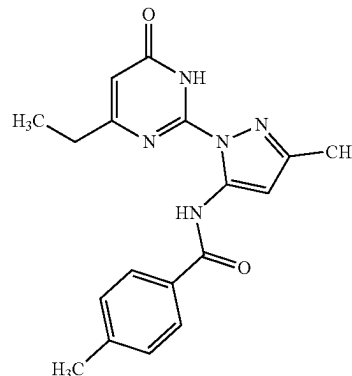
N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)-3-methylbenzamide (18)
AC10079



Prepared using general procedure A with 3-methylbenzoic acid (40.4 mg, 296 μ mol), BTFFH (108.0 mg, 341.6 μ mol), DCM (440 μ L), DIPEA (180 μ L, 1.0 mmol), and (4) (50.0 mg, 228 μ mol) to produce AC10079 (14.9 mg, 44.2 μ mol, 19.4%) as a white solid. ¹H NMR (800 MHz, Chloroform-d) δ 12.30 (s, 1H), 10.32 (s, 1H), 7.80-7.77 (m, 2H), 7.43-7.40 (m, 2H), 6.89 (s, 1H), 6.10 (d, J=0.9 Hz, 1H), 2.67 (q, J=7.5 Hz, 2H), 2.46 (s, 3H), 2.31 (s, 3H), 1.29 (t, J=7.6 Hz, 3H). ¹³C NMR (201 MHz, Chloroform-d) δ 168.02, 163.70, 160.93, 154.16, 148.75, 141.16, 138.83, 133.50, 133.17, 128.81, 127.76, 124.57, 107.67, 98.99, 30.70, 21.33, 14.23, 12.32. APCI-MS(+): m/z 338.2 [M+H], 198.1, 119.1. APCI-MS(-): m/z 336.1 [M-H]. HPLC retention time: 13.084 min. HPLC purity >99.0%.

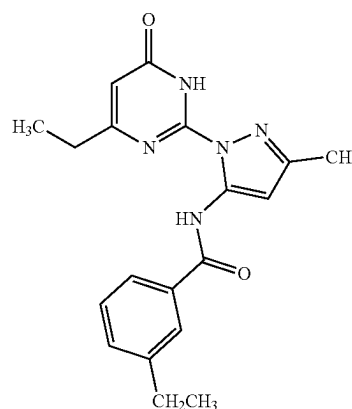
38

N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)-4-methylbenzamide (19)
AC10080



Prepared using general procedure with 4-methylbenzoic acid (40.4 mg, 296 μ mol), BTFFH (108.0 mg, 341.6 μ mol), DCM (440 μ L), DIPEA (180 μ L, 1.0 mmol), and (4) (50.0 mg, 228 μ mol) to produce AC10080 (32.8 mg, 97.2 μ mol, 42.6%) as a white solid. ¹H NMR (800 MHz, Chloroform-d) δ 12.31 (s, 1H), 10.31 (s, 1H), 7.87 (d, J=8.3 Hz, 2H), 7.32 (d, J=8.3 Hz, 2H), 6.87 (s, 1H), 6.10 (t, J=0.9 Hz, 1H), 2.66 (q, J=7.6 Hz, 2H), 2.45 (s, 3H), 2.31 (s, 3H), 1.31 (t, J=7.6 Hz, 3H). ¹³C NMR (201 MHz, Chloroform-d) δ 168.05, 163.49, 160.94, 154.16, 148.75, 143.49, 141.25, 130.39, 129.57, 127.33, 107.65, 98.87, 30.66, 21.59, 14.23, 12.32. APCI-MS(+): m/z 338.2 [M+H], 198.1, 119.1. APCI-MS(-): m/z 336.0 [M-H]. HPLC retention time: 13.039 min. HPLC purity 96.55%.

3-ethyl-N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)benzamide (20)
AC10081

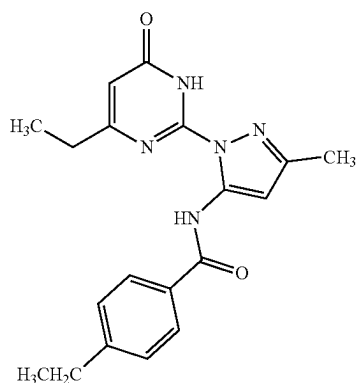


Prepared using general procedure with 3-ethylbenzoic acid (44.5 mg, 296 μ mol), BTFFH (108.0 mg, 341.6 μ mol), DCM (440 μ L), DIPEA (180 μ L, 1.0 mmol), and (4) (50.0 mg, 228 μ mol) to produce AC10081 (8.5 mg, 24 μ mol, 11%) as a white solid. ¹H NMR (800 MHz, Chloroform-d) δ 12.29 (s, 1H), 10.32 (s, 1H), 7.81 (d, J=2.0 Hz, 1H), 7.78 (dt, J=6.9, 1.9 Hz, 1H), 7.46-7.43 (m, 2H), 6.90 (s, 1H), 6.10 (d,

39

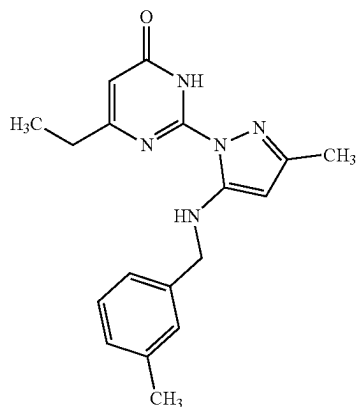
J=1.2 Hz, 1H), 2.75 (q, J=7.7 Hz, 2H), 2.66 (q, J=7.6 Hz, 2H), 2.31 (s, 3H), 1.30 (t, J=7.8 Hz, 3H), 1.28 (t, J=7.7 Hz, 3H). ¹³C NMR (201 MHz, Chloroform-d) δ 168.05, 163.80, 160.92, 154.15, 148.73, 145.29, 141.17, 133.26, 132.31, 128.85, 126.99, 124.59, 107.64, 99.03, 30.71, 28.88, 15.61, 14.23, 12.30. APCI-MS(+): m/z 352.2 [M+H], 212.1, 133.1. APCI-MS(-): m/z 350.1 [M-H]. HPLC retention time: 13.557 min. HPLC purity 96.96%.

4-ethyl-N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)benzamide (21)
AC10082



Prepared using general procedure A with 4-ethylbenzoic acid (44.5 mg, 296 μmol), BTFFH (108.0 mg, 341.6 μmol), DCM (440 μL), DIPEA (180 μL, 1.0 mmol), and (4) 50.0 mg, 228 μmol to produce AC10082 (20.1 mg, 57.2 μmol, 25.1%) as a white solid. ¹H NMR (800 MHz, Chloroform-d) δ 12.31 (s, 1H), 10.31 (s, 1H), 7.92-7.89 (m, 2H), 7.35 (d, J=8.0 Hz, 2H), 6.88 (s, 1H), 6.10 (d, J=0.9 Hz, 1H), 2.75 (q, J=7.7 Hz, 2H), 2.67 (q, J=7.5 Hz, 2H), 2.31 (s, 3H), 1.31 (t, J=7.6 Hz, 3H), 1.29 (t, J=7.7 Hz, 3H). ¹³C NMR (201 MHz, Chloroform-d) δ 168.07, 163.51, 160.93, 154.17, 149.69, 148.76, 141.26, 130.58, 128.39, 127.45, 107.65, 98.88, 30.67, 28.89, 15.23, 14.23, 12.33. APCI-MS(+): m/z 352.2 [M+H], 133.1. APCI-MS(-): m/z 350.1 [M-H]. HPLC retention time: 13.531 min. HPLC purity 98.10%.

6-ethyl-2-(3-methyl-5-((3-methylbenzyl)amino)-1H-pyrazol-1-yl)pyrimidin-4(1H)-one (22) AC10084

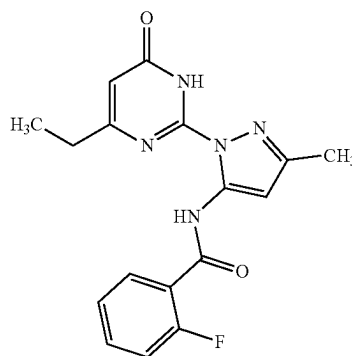


40

(4) (50.0 mg, 0.228 mmol, 1 eq.) was added to small borosilica vial equipped with stirbar. It was then dissolved with 250 μL of trifluoroacetic acid (TFA). Sodium triacetoxhydroborate (STAB) (242 mg, 1.14 mmol, 5 eq.) was added, followed by dropwise addition of a solution of 3-methylbenzaldehyde (40.3 μL, 0.342 mmol, 1.5 eq.) diluted in 250 μL of DCM. The mixture was allowed to stir overnight at room temperature.

The following morning, the reaction solution was concentrated on rotavap and loaded onto reverse phase flash column with THF. The flash column eluate was an ACN: H₂O gradient (20->95 ACN, H₂O containing 0.1% TFA). Fractions containing product were identified by UV and ESI MS and combined and concentrated on rotavap, yielding a white solid as product (3.7 mg, 11 μmol, 5.0%). ¹H NMR (800 MHz, DMSO-d₆) δ 11.56 (s, 1H), 8.14 (s, 1H), 7.23 (t, J=7.5 Hz, 1H), 7.19 (s, 1H), 7.16 (d, J=7.6 Hz, 1H), 7.08 (d, J=7.6 Hz, 1H), 6.01 (s, 1H), 4.34 (d, J=5.7 Hz, 2H), 2.29 (s, 3H), 2.10 (s, 3H), 1.10 (t, J=7.5 Hz, 3H). ¹³C NMR (201 MHz, DMSO-d₆) δ 152.02, 151.49, 138.58, 137.53, 128.31, 127.69, 127.54, 124.09, 87.07, 47.94, 20.91, 13.83, 11.87. ESI-MS(+): m/z 387.2 [M+CH₃CN+Na], 324.2 [M+H], 169.0, 126.1. HPLC retention time: 13.441 min. HPLC purity 96.95%

6-ethyl-2-(5-(2-fluorobenzyl)amino)-3-methyl-1H-pyrazol-1-yl)pyrimidin-4(1H)-one (23) AC10085

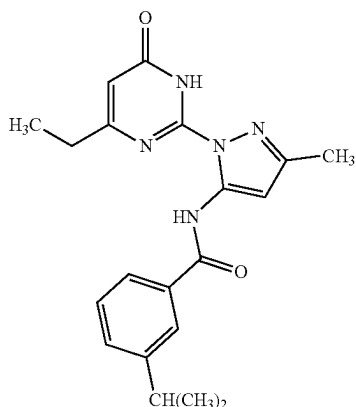


2-fluorobenzaldehyde (48.0 μL, 0.456 mmol, 3.96 eq.) and TFA (10.6 μL) were added to a small borosilica vial equipped with stirbar, followed by a small amount of 4 Å molecular sieves (activated overnight in oven). (4) (25.2 mg, 0.115 mmol, 1 eq.) was dissolved in 500 μL 1,1,1,3,3,3-hexafluoro-2-propanol (HFIP) and injected in portion to the stirring aldehyde mixture. The mixture was allowed to stir overnight at room temperature. To this mixture, 45 μL TFA and sodium cyanoborohydride (28.9 mg, 0.460 μmol, 4 eq.) were added sequentially. The reaction was allowed to stir 10 minutes at room temperature following cyanoborohydride addition. The reaction mixture was then filtered through a pad of celite to remove molecular sieves, diluted with 10 mL DCM and 10 mL potassium phosphate buffer (pH 7), and then transferred to a separatory flask. The mixture was then extracted thrice with 10 mL DCM. The organic layers were then collected, dried with MgSO₄, filtered, concentrated on rotavap, and loaded onto reverse phase column similar to (22). Fractions containing product were rotavapped to remove acetonitrile, after which the aqueous layer was neutralized with saturated sodium bicarbonate and then extracted thrice with an equal volume of DCM. The organic

41

layers were again dried and concentrated as previously. The resultant film was then washed with hexanes and diethyl ether to remove remaining DCM and concentrated, yielding an off-white powder as product (5.7 mg, 17 μ mol, 15%). ¹H NMR (800 MHz, DMSO-d₆) δ 11.46 (s, 1H), 8.17 (s, 1H), 7.45 (t, J=7.7 Hz, 1H), 7.34 (q, J=7.4 Hz, 1H), 7.22 (d, J=10.6 Hz, 1H), 7.18 (dt, J=7.7, 4.8 Hz, 1H), 6.03 (s, 1H), 5.41 (s, 1H), 4.44 (d, J=6.1 Hz, 2H), 2.52 (q, J=8.0 Hz, 2H), 2.10 (s, 3H), 1.11 (t, J=4.7 Hz, 3H). ¹³C NMR (201 MHz, DMSO-d₆) δ 160.93, 159.72, 151.19, 129.51, 129.49, 129.29, 129.25, 125.45, 125.38, 124.38, 115.27, 115.17, 87.00, 42d.06, 13.83, 11.81. ESI-MS(+): 391.1 [M+CH₃CN+Na], 328.2 [M+H], 169.0, 126.0. HPLC retention time: 12.961 min. HPLC purity 97.96%.

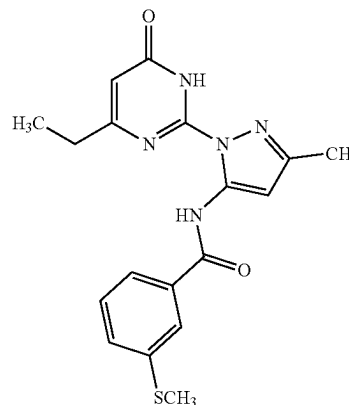
N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)-3-isopropylbenzamide (24)
AC10087



Prepared using general procedure A with 3-isopropylbenzoic acid (48.7 mg, 296 μ mol), BTFFH (108.0 mg, 341.6 μ mol), DCM (440 μ L), DIPEA (180 μ L, 1.0 mmol), and (4) (50.0 mg, 228 μ mol) to produce AC10087 (3.4 mg, 9.3 μ mol, 4.1%) as a white solid. ¹H NMR (800 MHz, Chloroform-d) δ 12.27 (s, 1H), 10.30 (s, 1H), 7.86 (t, J=1.9 Hz, 1H), 7.75 (dt, J=7.7, 1.5 Hz, 1H), 7.48 (dt, J=7.6, 1.5 Hz, 1H), 7.44 (t, J=7.6 Hz, 1H), 6.90 (s, 1H), 6.10 (s, 1H), 3.00 (hept, J=7.0 Hz, 1H), 2.66 (q, J=7.5 Hz, 2H), 2.32 (s, 3H), 1.31 (d, J=6.9 Hz, 6H), 1.28 (t, J=7.6 Hz, 3H). ¹³C NMR (201 MHz, Chloroform-d) δ 168.06, 163.94, 160.90, 154.15, 150.06, 148.71, 141.17, 133.29, 130.74, 128.79, 126.11, 124.40, 107.62, 99.05, 34.34, 30.73, 23.91, 14.24, 12.27. APCI-MS (+): m/z 366.2 [M+H], 226.1, 147.0. APCI-MS (-): m/z 364.1 [M-H], 131.5. HPLC retention time: 13.911 min. HPLC purity 95.94%.

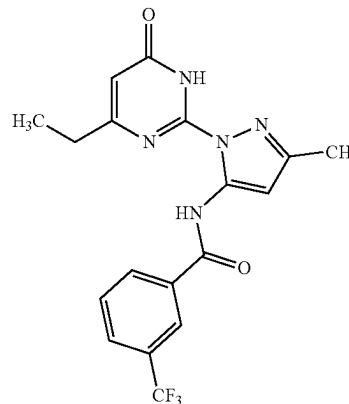
42

N-(1-(4-ethyl-6-oxo-1,6-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)-3-(methylthio)benzamide
(25) AC10088



Prepared using general procedure A with 3-(methylthio)benzoic acid (49.9 mg, 296 μ mol), BTFFH (108.0 mg, 341.6 μ mol), DCM (440 μ L), DIPEA (180 μ L, 1.0 mmol), and (4) (50.0 mg, 228 μ mol) to produce AC10088 (26.4 mg, 71.5 μ mol, 31.3%) as a white solid. ¹H NMR (800 MHz, Chloroform-d) δ 12.37 (s, 1H), 10.30 (s, 1H), 7.86 (td, J=1.8, 0.5 Hz, 1H), 7.71 (ddd, J=7.4, 1.7, 1.3 Hz, 1H), 7.46 (ddd, J=7.9, 1.9, 1.3 Hz, 1H), 7.45-7.43 (m, 1H), 6.88 (s, 1H), 6.10 (s, 1H), 2.69 (q, J=7.5 Hz, 2H), 2.55 (s, 3H), 2.31 (s, 3H), 1.29 (t, J=7.5 Hz, 3H). ¹³C NMR (201 MHz, Chloroform-d) δ 168.13, 163.08, 160.88, 154.12, 148.70, 140.98, 140.42, 133.77, 130.14, 129.25, 125.49, 123.56, 107.67, 99.07, 30.76, 15.77, 14.23, 12.28. APCI-MS(+): m/z 392.2 [M+Na], 370.2 [M+H], 230.1, 151.0. APCI-MS(-): m/z 390.0 [M+Na-2H], 368.1 [M-H]. HPLC retention time: 13.222 min. HPLC purity 97.27%.

N-(1-(4-ethyl-6-oxo-1,6-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)-3-(trifluoromethyl)benzamide (26) AC10089

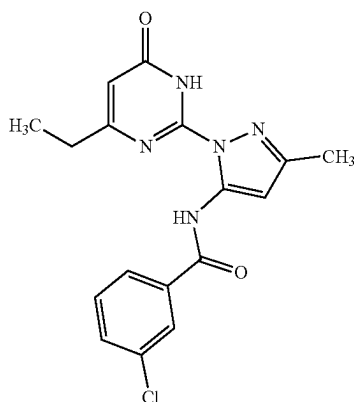


Prepared using general procedure A with 3-(trifluoromethyl)benzoic acid (56.4 mg, 296 μ mol), BTFFH (108.0 mg, 341.6 μ mol), DCM (440 μ L), DIPEA (180 μ L, 1.0 mmol), and (4) (50.0 mg, 228 μ mol) to produce AC10089 (20.5 mg,

43

52.4 μ mol, 23.0%) as a white solid, despite spilling a portion during work-up. ^1H NMR (800 MHz, Chloroform- d) δ 12.45 (s, 1H), 10.33 (s, 1H), 8.22 (d, $J=7.9$ Hz, 1H), 8.18 (s, 1H), 7.88 (d, $J=7.9$ Hz, 1H), 7.70 (t, $J=7.9$ Hz, 1H), 6.92 (s, 1H), 6.11 (s, 1H), 2.65 (q, $J=7.6$ Hz, 2H), 2.33 (s, 3H), 1.25 (t, $J=7.5$ Hz, 3H). ^{13}C NMR (201 MHz, Chloroform- d) δ 168.04, 162.01, 160.81, 154.12, 148.71, 140.66, 134.12, 131.16, 129.74, 129.29, 123.68 (d, $J=3.8$ Hz), 107.71, 99.48, 30.59, 14.23, 12.06. APCI-MS(+): m/z 392.2 $[\text{M}+\text{H}]$, 252.1, 220.1. APCI-MS(-): m/z 390.0 $[\text{M}-\text{H}]$. HPLC retention time: 13.454 min. HPLC purity 96.54%.

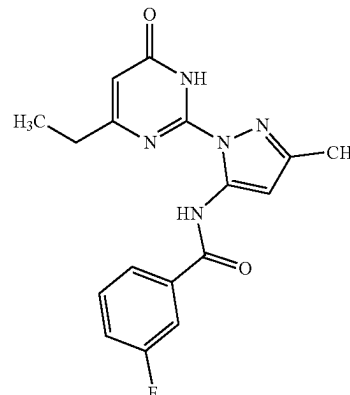
3-chloro-N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)benzamide (27)
AC10090



Prepared using general procedure A with 3-chlorobenzoic acid (46.4 mg, 296 μ mol), BTFFH (108.0 mg, 341.6 μ mol), DCM (440 μ L), DIPEA (180 μ L, 1.0 mmol), and (4) (50.0 mg, 228 μ mol) to produce AC10090 (19.0 mg, 53.1 μ mol, 23.3%) as a white solid. ^1H NMR (800 MHz, Chloroform- d) δ 12.45 (s, 1H), 10.30 (s, 1H), 7.93-7.90 (m, 2H), 7.59 (d, $J=8.0$ Hz, 1H), 7.49 (t, $J=7.8$ Hz, 1H), 6.88 (s, 1H), 6.11 (s, 1H), 2.70 (q, $J=7.5$ Hz, 2H), 2.32 (s, 3H), 1.30 (t, $J=7.5$ Hz, 3H). ^{13}C NMR (201 MHz, Chloroform- d) δ 168.03, 162.02, 160.84, 154.15, 148.75, 140.79, 135.13, 134.91, 132.72, 130.38, 127.05, 125.94, 107.77, 99.20, 30.74, 14.23, 12.32. APCI-MS(+): m/z 360.1, 358.1 $[\text{M}+\text{H}]$, 220.0, 218.0, 140.8, 138.9. APCI-MS(-): m/z 357.9, 355.9 $[\text{M}-\text{H}]$. HPLC retention time: 13.470 min. HPLC purity 98.08%.

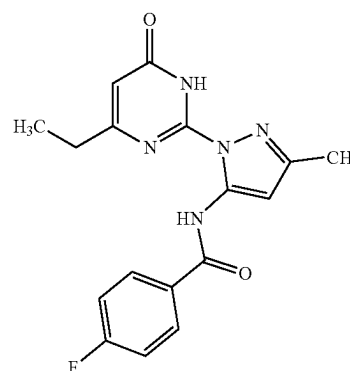
44

N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)-3-fluorobenzamide (28)
AC10091



Prepared using general procedure A with 3-fluorobenzoic acid (46.4 mg, 296 μ mol), BTFFH (108.0 mg, 341.6 μ mol), DCM (440 μ L), DIPEA (180 μ L, 1.0 mmol), and (4) (50.0 mg, 228 μ mol) to produce AC10091 (8.8 mg, 26 μ mol, 11%) as a white solid. Required additional purification via normal phase flash chromatography (DCM:MeOH, 10% MeOH isocratic). ^1H NMR (800 MHz, Chloroform- d) δ 12.43 (s, 1H), 10.29 (s, 1H), 7.78 (ddd, $J=7.7, 1.7, 1.0$ Hz, 1H), 7.68 (ddd, $J=9.2, 2.6, 1.6$ Hz, 1H), 7.52 (td, $J=8.0, 5.4$ Hz, 1H), 7.32 (tdd, $J=8.2, 2.6, 0.9$ Hz, 1H), 6.88 (s, 1H), 6.11 (s, 1H), 2.66 (q, $J=7.5$ Hz, 2H), 2.32 (s, 3H), 1.30 (t, $J=7.6$ Hz, 3H). ^{13}C NMR (201 MHz, Chloroform- d) δ 167.94, 163.57, 162.33, 162.14, 160.80, 154.17, 140.81, 135.45, 130.67 (d, $J=7.9$ Hz), 122.96, 119.77 (d, $J=21.1$ Hz), 114.50 (d, $J=23.1$ Hz), 107.83, 99.18, 30.65, 14.23, 12.31. APCI-MS(+): 342.1 $[\text{M}+\text{H}]$, 202.0, 123.0. APCI-MS(-): 340.0 $[\text{M}-\text{H}]$. HPLC retention time: 12.844 min. HPLC purity >99.0%.

N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)-4-fluorobenzamide (29)
AC10092

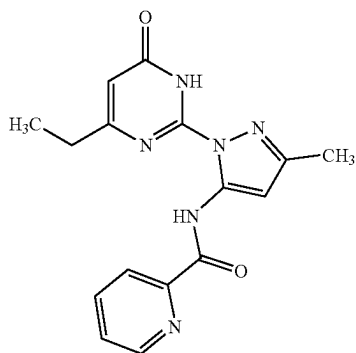


Prepared using general procedure A with 4-fluorobenzoic acid (46.4 mg, 296 μ mol), BTFFH (108.0 mg, 341.6 μ mol), DCM (440 μ L), DIPEA (180 μ L, 1.0 mmol), and (4) (50.0 mg, 228 μ mol) to produce AC10092 (19.6 mg, 57.4 μ mol, 25.2%) as a white solid. ^1H NMR (800 MHz, Chloroform- d)

45

δ 12.34 (s, 1H), 10.30 (s, 1H), 8.01-7.98 (m, 2H), 7.23-7.19 (m, 2H), 6.87 (s, 1H), 6.11 (s, 1H), 2.64 (q, $J=7.6$ Hz, 2H), 2.31 (s, 3H), 1.29 (t, $J=7.5$ Hz, 3H). ^{13}C NMR (201 MHz, Chloroform- d) δ 167.81, 166.09, 164.82, 162.43, 160.81, 154.19, 148.78, 141.00, 129.73 (d), 116.10 (d), 107.78, 99.03, 30.66, 14.23, 12.26. APCI-MS(+): 342.1 $[\text{M}+\text{H}]$, 202.0, 123.0. APCI-MS(-): 340.0 $[\text{M}-\text{H}]$. HPLC retention time: 12.722 min. HPLC purity >99.0%.

N-(1-(6-ethyl-4-oxo-1,4-dihydropyrimidin-2-yl)-3-methyl-1H-pyrazol-5-yl)picolinamide (30) AC10093



Prepared using general procedure A with picolinic acid (36.5 mg, 296 μmol), BTFFH (108.0 mg, 341.6 μmol), DCM (440 μL), DIPEA (180 μL , 1.0 mmol), and (4) (50.0 mg, 228 μmol) to produce AC10093 (36.1 mg, 111 μmol , 48.8%) as a tan solid. ^1H NMR (800 MHz, Chloroform- d) δ 13.48 (s, 1H), 10.26 (s, 1H), 8.64 (ddd, $J=4.7, 1.7, 0.9$ Hz, 1H), 8.28 (dt, $J=7.8, 1.1$ Hz, 1H), 7.94 (td, $J=7.7, 1.7$ Hz, 1H), 7.53 (ddd, $J=7.5, 4.7, 1.2$ Hz, 1H), 6.93 (s, 1H), 6.11 (s, 1H), 2.77 (q, $J=7.6$ Hz, 2H), 2.32 (s, 3H), 1.43 (t, $J=7.6$ Hz, 3H). ^{13}C NMR (201 MHz, Chloroform- d) δ 169.11, 161.86, 161.31, 153.78, 148.92, 148.27, 148.21, 140.61, 137.68, 127.05, 122.96, 107.66, 99.00, 30.71, 14.23, 12.33. APCI-MS(+): m/z 325.1 $[\text{M}+\text{H}]$, 281.1, 185.0. APCI-MS(-): m/z 323.0 $[\text{M}-\text{H}]$. HPLC retention time: 12.473 min. HPLC purity 95.10%.

Those skilled in the art will recognize that numerous modifications can be made to the specific implementations described above. The implementations should not be limited to the particular limitations described. Other implementations may be possible.

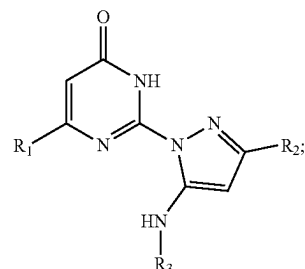
While the inventions have been illustrated and described in detail in the drawings and foregoing description, the same is to be considered as illustrative and not restrictive in character, it being understood that only certain embodiments have been shown and described and that all changes and modifications that come within the spirit of the invention are desired to be protected.

It is intended that that the scope of the present methods and compositions be defined by the following claims. However, it must be understood that this disclosure may be practiced otherwise than is specifically explained and illustrated without departing from its spirit or scope. It should be understood by those skilled in the art that various alternatives to the embodiments described herein may be employed in practicing the claims without departing from the spirit and scope as defined in the following claims.

46

What is claimed is:

1. A method for treatment of adenylyl cyclase-1 (AC-1) mediated pain or opioid dependence thereof, which comprises administering a therapeutically effective amount of a compound of Formula (I):



wherein:

R_1 is a $\text{C}_1\text{-C}_6$ alkyl;

R_2 is a $\text{C}_1\text{-C}_6$ alkyl; and

R_3 is an acyl, alkyl, alkenyl, alkynyl, heteroalkyl, heteroalkenyl, heteroalkynyl, cycloalkyl, heterocycloalkyl, cycloalkenyl, heterocycloalkenyl, heterocyclyl; or an optionally substituted aryl, arylalkyl, or arylalkenyl;

wherein:

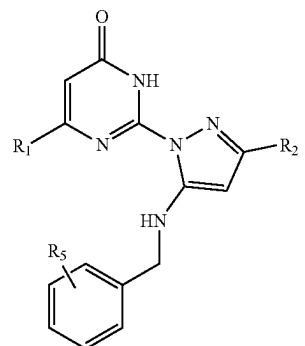
the compound of Formula (I) is a adenylyl cyclase-1 (AC-1) inhibitor:

the optionally substituted aryl, arylalkyl, or arylalkenyl, respectively, is further optionally substituted with optional substituents selected from a halogen group; hydroxyl group; alkoxy group; aryloxy group; aralkyloxy group; oxo(carbonyl) group; carboxylic acid group; carboxylate group; carboxylate ester group; sulfur atom containing groups selected from thiol, thioalkyl, aryl sulfide, sulfoxide, sulfone, sulfonyl or sulfonamide groups; or nitrogen atom containing groups selected from amines, azides, hydroxylamines, cyano, nitro groups, N-oxides, hydrazides, or enamines; or

a pharmaceutically acceptable salt thereof to a subject in need thereof.

2. The method according to claim 1, wherein R_1 is ethyl and R_2 is methyl.

3. The method according to claim 1, wherein said compound has a Formula (V):



47

wherein:

R₁ is a C₁-C₆ alkyl;

R₂ is a C₁-C₆ alkyl; and

R₅ represents five substituents groups on phenyl ring of Formula (V):

wherein:

each one of those five substituent groups of R₅ independently is a substituent selected from the group consisting of hydrogen, deuterium, halo, azido, cyano, nitro, hydroxy, amino, thio, carboxy, ester, amide, and acyl, sulfoxyl, sulfonyl, phosphate, phosphoryl, alkyl, alkenyl, alkynyl, heteroalkyl, heteroalkenyl, heteroalkynyl, heterocyclyl, cycloalkyl, cycloalkenyl, cycloheteroalkyl, cycloheteroalkenyl, aryl, heteroaryl, arylalkyl, arylalkenyl, and arylalkynyl, each of which is

optionally substituted; or
any two adjacent substituents defined as optional substituents of R⁵ are taken together with the attached carbons to form an optionally substituted cyclic or heterocyclic moiety;

48

wherein:

each one of those five substituent groups of R₅, the optionally substituted cyclic or heterocyclic moiety, respectively, is further optionally substituted with optional substituents selected from a halogen group; hydroxyl group; alkoxy group; aryloxy group; aralkyloxy group; oxo(carbonyl) group; carboxylic acid group; carboxylate group; carboxylate ester group; sulfur atom containing groups selected from thiol, thioalkyl, aryl sulfide, sulfoxide, sulfone, sulfonyl or sulfonamide groups; or nitrogen atom containing groups selected from amines, azides, hydroxylamines, cyano, nitro groups, N-oxides, hydrazides, or enamines; or

a pharmaceutically acceptable salt thereof to a subject in need thereof.

* * * * *